SYPT 2027 · PROBLEM B8 · COLD DRINK

# The Salted Ice Bath

The physics of the bath and the drink, from first principles

Handover report — October 2026 · built from the Drive file `sypt code`

---

A can of drink is cooled in a bath of crushed ice, water and salt. This report derives everything that decides how cold the bath gets and how fast the drink cools, starting from conservation of energy and of mass, the thermodynamics of salt water, and the laws of conduction, convection and diffusion. Every equation is derived where it is first used, its symbols are named with their units, and the conditions it needs are stated. Where physics gives only the form of a law and experiments supply its constant (the freezing curve of strong brine, free-convection correlations, the permeability of a heap of ice), the report says so and gives the source.

Two programs put the equations to work. `b8_sim.py` treats the bath as one body at equilibrium and answers how cold (§12). `b8_model.py`, behind the interactive simulator, follows every process at its own rate and answers how fast (§13). The colour code of the reporter deck is kept: **FLOOR** for equilibrium thermodynamics (how cold) and **RATE** for heat transfer (how fast).

## Contents

1 Summary and status 2

2 Starting point: what `sypt code` did 3

Part I · The physics 3

3 The system, and what is conserved 3

4 Salt water: freezing, solubility and the eutectic 5

5 The bath at equilibrium: how cold 7

6 How heat moves 9

7 Melting, dissolving and freezing: the rates 11

8 Where the drink's heat goes: spreading through the bath 12

9 The liquid level and the areas that follow it 14

10 The box: heat from the room 15

11 Putting it together: two models and how they are solved 17

Part II · What the models say 18

12 How cold: results of the equilibrium model 18

13 How fast: results of the simulator 22

Part III · What to do next 23

14 Open problem: why did the one-shot bath stall? 23

15 Experimental programme 24

16 Estimates justifying the approximations 24

17 Literature 25

18 Notes for whoever picks this up 26

A Symbols 27

B Using the code 27

# 1 Summary and status

**What the physics says.** The temperature of the bath is not fixed by its recipe alone but by its energy: the brine composition depends on how much ice has melted, and that depends on the temperature (§2). Closing that loop with an energy balance gives four results that change the rig. **The charge in the rig note does not reach the eutectic**: it settles near $-17.8\,^{\circ}$C with no salt left, so it has no thermostat (§12.7). **4.0 kg of ice, 1.6 kg of water and 1.0 kg of salt** reaches $-21.2\,^{\circ}$C and keeps 193 g of salt as a buffer. Reaching the eutectic is an energy condition, not a mass condition (§5.5). And the drink's heat does not reach the whole bath at once: it spreads through the ice bed no faster than the brine can carry it (§8). With the can lying on its side as in the rig, the simulator puts Can 2 at 5 °C **222 s** after it goes in (§13).

**Solid.** The enthalpy formulation and the equilibrium solve it implies (§5); the reproduction of every bath number in the earlier B8 notes once their starting temperatures are matched (Table 2); the result that the bath temperature cannot be found from composition alone; the $1/$size scaling of bath drift; the condition for reaching the eutectic; the numerical methods, which match closed-form solutions and conserve energy to $10^{-9}$ J.

**Computed but not measured.** The adiabatic temperature of any charge; the buffer map (Fig. 5); the drink cooling curves for Can 2; the blank-run warming rate; how far the drink's heat spreads; the box's heat leak.

**Uncertain.** The heat-transfer coefficients across the can, taken from a bottle study and a SINTEF time constant rather than measured on this rig: every **RATE** number scales with them. The effective conductivity of the ice bed, an estimate good to a factor of two or three. The rates at which ice melts and salt dissolves. The **FLOOR** results move by at most 0.3 K over the plausible range of the textbook constants (§16).

**Not attempted.** Layering of the bath from top to bottom; the metastable eutectic and hydrohalite nucleation; flow inside the drink (the COMSOL model covers that); and any dissolution law that could explain the bench stall at $-7.5\,^{\circ}$C (§14).

## 2 Starting point: what `sypt code` did

The script defines the right ingredients — 1 kg ice, 2 kg water, 0.5 kg salt, $L_f = 333.55\,\mathrm{kJ\,kg^{-1}}$, $K_f = 1.86\,\mathrm{K\,kg\,mol^{-1}}$, $i = 2$, and a linear solubility fit from the Sigma-Aldrich table — and stops. Read bluntly:

| Line | Problem, and what it costs |
|---|---|
| `bathT = 25.0 ## K` | It is 25 °C. 25 K is $-248\,^\circ$C. Every temperature in the model is Celsius; the label should say so. |
| `solubility()` | Compares `paramsaltMass` in kg with a fit in *grams per 100 g of water*. 0.5 is always less than 35.2, so the cap can never bind, for any input. Correct form: $s_{\max} = 10^{-2}\,(0.0338\,T + 35.24)\,m_w$ kg. The fit is also the *halite* branch, valid above $+0.1\,^\circ$C; below that the solid in equilibrium is hydrohalite and the solubility falls to 30.2 g/100 g at the eutectic, not the 34.5 the fit extrapolates to. |
| `temp()` | Computes $\Delta T_f$ and returns nothing, so the function evaluates to `None`. |
| molality in `temp()` | Divides by the initial 2 kg of water. The liquid water at equilibrium is 2 kg *plus whatever ice melted*, and that is the unknown. |
| Blagden, $i = 2$ | Over-predicts the depression by 11 % at 5 wt %, under-predicts by 9 % near the eutectic (§4.2), and has no floor: at 30 wt % it gives $-27.3\,^\circ$C for a brine that cannot exist. |
| `1.201 Density ... as a %` | Density is in g mL$^{-1}$, not a percentage. Unused, but `waterVol` is used as a mass: fine for water, 20 % wrong for brine. |
| `import matplotlib, time` | `pyplot` is never imported; `main()` times nothing. Harmless. |

The one that matters is structural, not a bug. **The temperature depends on the brine composition, the composition depends on how much ice has melted, and how much ice melts depends on the temperature.** Composition alone cannot close that loop; an energy balance can. For the script's own charge, `temp()` would have reported $-15.9\,^\circ$C (20 wt %, all salt in the original 2 kg). The equilibrium is $-13.7\,^\circ$C with 647 g of ice left if the water starts at 0 °C, and $-10.5\,^\circ$C with only 51 g of ice left if it starts at 25 °C — a bath one cold drink away from running out of ice. Everything below is the loop closed.

# Part I · The physics

## 3 The system, and what is conserved

### 3.1 The parts

The rig is a can of drink standing (or lying) in a glass tank, inside a styrofoam box with a lid, in a room at $T_{\text{room}} = 28\,^\circ$C. The tank holds the bath: liquid *brine* (water with salt dissolved in it), crushed *ice*, and undissolved *salt crystals* on the floor. Each part holds heat and passes it to its neighbours:

$$\text{room} \to \text{foam} \to \text{glass} \to \text{brine}, \qquad \text{room} \to \text{foam lid} \to \text{air} \to \text{brine},$$
$$\text{drink} \to \text{can wall} \to \text{brine} \to \text{ice}.$$

Two quantities are conserved in every part and every exchange: energy and mass. Everything that follows is these two conservation laws, written for each part, together with the laws that say how

fast energy and mass move between parts.

## 3.2 Energy: the first law at constant pressure

For any body that does no work other than by changing its volume against the surrounding pressure $p$, the first law of thermodynamics reads

$$\mathrm{d}U = \delta Q - p\,\mathrm{d}V, \tag{1}$$

with $U$ the internal energy (J), $\delta Q$ the heat received (J) and $V$ the volume ($\mathrm{m}^3$). The tank is open to the room, so $p$ stays at one atmosphere. Defining the *enthalpy* $H = U + pV$,

$$\mathrm{d}H = \mathrm{d}U + p\,\mathrm{d}V + V\,\mathrm{d}p = \delta Q \qquad (\mathrm{d}p = 0). \tag{2}$$

*Where it comes from.* Equation (1) is energy conservation: what a body gains as heat it keeps as internal energy, less the work it does pushing the atmosphere back as it expands. Adding $pV$ folds that work into the bookkeeping, so at constant pressure every joule of heat that arrives shows up as enthalpy. Ice shrinks by 8 % as it melts and brine expands slightly as it warms; with $H$ these volume changes need no separate treatment.

*When it holds.* At constant pressure, with no stirring work counted as heat (a stirrer's power $P_{\mathrm{stir}}$ is added to the heat it stirs into).

For a part $k$ receiving heat flows $\dot{Q}_{jk}$ (W) from its neighbours $j$,

$$\frac{\mathrm{d}H_k}{\mathrm{d}t} = \sum_j \dot{Q}_{jk}, \qquad \dot{Q}_{jk} = -\dot{Q}_{kj}. \tag{3}$$

Because every flow between two parts appears once with each sign, adding (3) over all the parts inside the box cancels them: the total enthalpy changes only by the heat that crosses the box from the room, plus the stirrer. The simulator checks exactly this after every run (the *energy bookkeeping*, §11).

## 3.3 Mass: water and salt

Water is present as ice (mass $m_i$) and as liquid in the brine ($m_l$); salt is present dissolved ($s$) and as crystals ($S - s$). Melting moves water from one form to the other and dissolving moves salt, but neither is created:

$$m_i + m_l = W, \qquad s + (S - s) = S, \qquad w = \frac{s}{m_l + s}, \tag{4}$$

with $W$ and $S$ the totals put in (kg) and $w$ the mass fraction of salt in the brine (kg of salt per kg of brine). A brine of $w = 0.232$ is 23.2 wt % salt.

## 3.4 The enthalpy of each phase

Enthalpy is defined up to a constant, so a reference state is chosen: **liquid water and solid salt at 0 °C have zero enthalpy.** Then, for each kilogram, from (2) with $\delta Q = c\,\mathrm{d}T$:

**Liquid water.** Warming from 0 °C to $T$ at constant pressure takes $c_w T$, with $c_w$ the specific heat capacity (J kg$^{-1}$K$^{-1}$): $h_w(T) = c_w T$.

**Ice.** Start from the reference, liquid water at 0 °C. Freezing it releases the latent heat, $L_0 = 333.55$ kJ per kg; cooling the ice to $T$ then takes out $\int c_i \,\mathrm{d}T$:

$$h_i(T) = -L_0 + \int_0^T c_i(T')\,\mathrm{d}T' \;\approx\; c_i T - L_0. \tag{5}$$

The heat that melts 1 kg of ice at a temperature $T$ below zero is then the difference of the two enthalpies at that temperature,

$$L(T) = h_w(T) - h_i(T) = L_0 + (c_w - c_i)\,T, \tag{6}$$

Kirchhoff's law. Because $c_w > c_i$, melting below 0 °C takes *less* heat: 287 kJ kg$^{-1}$ at $-21.2\,^\circ$C instead of 334. The simulator uses the IAPWS heat capacity of ice, which falls from 2097 J kg$^{-1}$K$^{-1}$ at 0 °C to 1950 at $-20\,^\circ$C ($c_i = c_{i0} + c_i' T$, so the integral in (5) gives $c_{i0}T + \frac{1}{2}c_i' T^2$); `b8_sim.py` holds $c_i$ constant.

**Salt.** Solid salt warms like any solid: $h_s = c_s T$, $c_s = 864$ J kg$^{-1}$K$^{-1}$ (50.5 J mol$^{-1}$K$^{-1}$ divided by 58.44 g mol$^{-1}$).

**Dissolved salt.** Dissolving NaCl in water absorbs heat: breaking the crystal lattice costs more than hydrating the ions returns. The heat of solution is $\Delta h_s = 3.88$ kJ mol$^{-1}$, or 66.4 kJ per kg of salt (measured at 25 °C and infinite dilution; it is held constant). A kilogram of salt that dissolves therefore carries $\Delta h_s$ more enthalpy than when it was a crystal, and the brine, which has to supply it, cools.

**The whole bath.** Adding the parts, with the brine (liquid water $m_l$ and dissolved salt $s$) taken to warm with the heat capacity $c_b(w)$ of brine of its composition:

$$H = \underbrace{(m_l + s)\,c_b(w)\,T}_{\text{brine, sensible}} + \underbrace{s\,\Delta h_s}_{\text{dissolution}} + \underbrace{m_i\,(c_i T - L_0)}_{\text{ice}} + \underbrace{(S - s)\,c_s T}_{\text{undissolved salt}}. \tag{7}$$

*Where it comes from.* Equation (7) is the sum of the per-kilogram enthalpies above, each multiplied by its mass, with every phase at the same temperature $T$. It is the single equation that connects the heat put into the bath with its temperature and phase amounts.

*When it holds.* When the bath is at one temperature (relaxed in the simulator, which gives every ring of brine and its ice their own temperatures, §7), when $\Delta h_s$ does not depend on concentration (±20 % moves the bath by less than 0.05 K), and when mixing water with salt changes the heat capacity but not the enthalpy otherwise.

The heat capacity of brine falls as salt is added. `b8_sim.py` uses $c_b = 4180 - 4400\,w$; the simulator uses a fit to Melinder (2010), $c_b = 4189 - 5794\,w + 8579\,w^2$ J kg$^{-1}$K$^{-1}$, good to $\pm 33$ between $-20$ and $+10\,^\circ$C. Table 1 lists every constant.

# 4 Salt water: freezing, solubility and the eutectic

## 4.1 Why salt lowers the freezing point

Ice and liquid water coexist when a molecule of water has no reason to move from one to the other: when the *chemical potential* of water, the Gibbs energy per mole, is the same in both. For pure water

**Table 1.** Constants. The first block is used by `b8_sim.py`; the simulator reads the same quantities, with the fits shown where they differ, from `inputs.toml`.

| Constant | Value | Unit | Source |
|---|---|---|---|
| $L_0$ (ice, 0 °C) | 333.55 | kJ kg$^{-1}$ | handbook |
| $c_w$, $c_i$, $c_s$ | 4.18, 2.00, 0.864 | kJ kg$^{-1}$K$^{-1}$ | $c_i$: simulator uses IAPWS R10-06 |
| $c_b(w)$ | $4.18 - 4.4\,w$ | kJ kg$^{-1}$K$^{-1}$ | simulator: Melinder fit |
| $\rho_b$ | 1180 | kg m$^{-3}$ | eutectic brine; simulator: $1000 + 786\,w$ (Melinder) |
| $\Delta h_s$ | $+66.4$ | kJ kg$^{-1}$ | 3.88 kJ mol$^{-1}$, infinite dilution, 25 °C |
| $T_E$, $w_E$ | $-21.2$, 0.2318 | °C, – | end of the Bodnar fit |
| $T_P$, $w_P$ | $+0.1$, 0.26285 | °C, – | Swenne (1983) peritectic |
| $k_b(w)$ | $0.5725 - 0.149\,w$ | W m$^{-1}$K$^{-1}$ | Melinder fit, simulator only |
| $C_d$ | 1.332 | kJ K$^{-1}$ | 330 g drink at $c = 4.0$ + 13 g Al |
| $U_0$, $A$ | 160, 0.03102 | W m$^{-2}$K$^{-1}$, m$^2$ | SINTEF $\tau \approx 265$ s; Can 2 |
| $(UA)_{\text{box}}$, $T_{\text{room}}$ | 0.46, 28 | W K$^{-1}$, °C | rig note, 30 mm EPS; simulator 0.34 |

this happens at $T_0 = 273.15$ K. In brine, dissolved ions dilute the water, which lowers its chemical potential by $RT \ln a_w$, with $a_w$ the activity of water ($a_w = 1$ for pure water, $a_w < 1$ in brine). Ice contains no salt, so its chemical potential is unchanged. Equilibrium at a new temperature $T$ needs

$$\mu_{\text{ice}}(T) = \mu_w^*(T) + RT \ln a_w, \tag{8}$$

with $\mu_w^*$ the chemical potential of pure liquid water and $R = 8.314$ J mol$^{-1}$K$^{-1}$. Dividing by $RT$ and using the Gibbs–Helmholtz relation $\mathrm{d}(\mu/T)/\mathrm{d}T = -h_m/T^2$ for each phase ($h_m$ the molar enthalpy), the temperature derivative of $\ln a_w$ is the molar latent heat $L_m = 6.01$ kJ mol$^{-1}$ over $RT^2$; integrating from $T_0$, where $a_w = 1$,

$$\ln a_w = -\frac{L_m}{R}\left(\frac{1}{T} - \frac{1}{T_0}\right). \tag{9}$$

For a dilute, ideal solution the activity is the mole fraction of water, so $\ln a_w = \ln(1 - x) \approx -x$, and $x \approx M_w\, i\, b$, with $b$ the molality (moles of NaCl per kg of water), $i = 2$ the number of ions each formula unit gives, and $M_w = 0.018015$ kg mol$^{-1}$. For a small depression $\Delta T = T_0 - T$, $1/T - 1/T_0 \approx \Delta T/T_0^2$. Together:

$$\Delta T = K_f\, i\, b, \qquad K_f = \frac{R\,T_0^2\,M_w}{L_m} = 1.86 \text{ K kg mol}^{-1}, \qquad b = \frac{w}{M_{\text{NaCl}}(1 - w)}. \tag{10}$$

This is Blagden's law, the one `sypt code` used, and its constant follows from the latent heat alone.

*When it holds.* Only when the brine is dilute (a few per cent), ideal (no ion–ion interactions), and fully dissociated, and when $\Delta T \ll T_0$. None of these is true near the eutectic, where $b = 5.2$ mol kg$^{-1}$ and one particle in six is an ion.

## 4.2 The measured freezing curve

Strong brine is not ideal: water molecules bound in the ions' hydration shells are not free to freeze, and the ions screen each other. Thermodynamics still gives (9) exactly; what fails is the step $a_w \approx 1 - M_w i b$. Rather than model the activity, the model uses the measured freezing curve, the *liquidus*, in the form fitted by Bodnar (1993) to fluid-inclusion data, with $\theta = -T$ the depression below 0 °C:

$$w_L(T) = 0.0178\,\theta - 4.42 \times 10^{-4}\,\theta^2 + 5.572 \times 10^{-6}\,\theta^3, \qquad 0 \le \theta \le 21.2. \tag{11}$$

*Where it comes from.* Experiment, with a check from (10). For dilute brine, (10) gives $w \approx 0.0157\,\theta$, since $M_{\mathrm{NaCl}}/(iK_f) = 0.0157\,\mathrm{K}^{-1}$: close to the leading term of the fit. The higher terms carry the non-ideality. The ratio of Blagden's depression to the measured one is 1.106, 1.069, 1.019, 0.952 and 0.909 at 5, 10, 15, 20 and 23 wt % (Fig. 1). The fit is explicit in $T$, which makes the equilibrium solve fast; the simulator inverts it by Newton's method when it needs $T$ from $w$.

### 4.3 Solubility, the eutectic and the phase rule

How many of the bath's variables are free is set by Gibbs' phase rule. At fixed pressure, a mixture of $C$ components in $P$ phases at equilibrium has

$$F = C - P + 1 \tag{12}$$

free intensive variables: each phase has $C - 1$ independent mass fractions, the temperature adds one, and each of the $C(P-1)$ equalities of chemical potential between phases removes one. With water and salt ($C = 2$):

- brine alone ($P = 1$, $F = 2$): temperature and composition are both free;
- brine with ice, or brine with salt crystals ($P = 2$, $F = 1$): composition follows from temperature — the liquidus (11), or the solubility line $w_{\mathrm{sat}}(T)$;
- brine with ice *and* salt ($P = 3$, $F = 0$): nothing is free. This is the *eutectic* point, $T_E = -21.2\,^{\circ}\mathrm{C}$, $w_E = w_L(T_E) = 0.2318$.

On the eutectic the bath is a thermostat: heat put in cannot change the temperature, only the amounts of the three phases. The solubility of halite (rock salt) is $35.65 + 0.0147\,T$ g per 100 g of water between 0 and 30 °C, $w_{\mathrm{sat}} = g/(100+g)$. Below the peritectic, $T_P = +0.1\,^{\circ}\mathrm{C}$, the stable solid is hydrohalite NaCl·2$H_2$O, whose solubility falls from $w_P = 0.2628$ to $w_E$ at the eutectic (taken as a straight line). Brine on the liquidus is below both solubility lines everywhere above $T_E$, so **crystals and ice can coexist only at the eutectic**: with ice present, salt dissolves until it is all gone or the brine reaches $w_E$.

## 5 The bath at equilibrium: how cold

### 5.1 One idea: describe the bath by its enthalpy

Given the totals $W$ and $S$ and the enthalpy $H$, the bath has exactly one equilibrium state. The temperature is then an output, not an input: heat that crosses the bath boundary changes $H$ (3), and temperature, ice, brine composition and salt left on the floor are all read off $H$. This removes the circularity of §2, and five behaviours that are usually bolted on as separate rules come out of one function: the adiabatic mixing temperature, the drift along the liquidus as ice melts, the eutectic thermostat, the exhaustion of the ice, and the temperature dependence of the latent heat.

### 5.2 The equilibrium solve $T_b = \mathcal{T}(H;\, W, S)$

Combining the mass balances (4), the phase rule (§4.3) and the enthalpy (7) gives four regimes:

| Regime | Phase amounts (from mass and phase equilibrium) | $T_b$ |
|---|---|---|
| `liquid` | no ice, $m_l = W$; $s = \min(S,\ W\, r_{\text{sat}}(T))$ | from (7) |
| `liquidus` | $w = w_L(T)$, $s = S$, $m_l = S\,(1 - w_L)/w_L$, $m_i = W - m_l$ | from (7) |
| `eutectic` | $w = w_E$, $0 \le m_l \le \min(W, S/r_E)$, $s = r_E\, m_l$ | $T_E$, fixed |
| `solid` | no liquid | below $T_E$ |

Here $r = w/(1-w)$ is kilograms of salt per kilogram of water, and $r_E = 0.3018$ at the eutectic.

*Where it comes from.* `liquidus`: with ice present the brine sits on the freezing curve ($F = 1$); all salt is dissolved because the liquidus lies below saturation (§4.3); the salt balance $s = w\,(m_l + s)$ then gives $m_l$, and the water balance gives $m_i$. `eutectic`: three phases, $F = 0$, so $T$ and $w$ are fixed and only the amount of brine is unknown; (7) is linear in $m_l$ there, so the solve is a single division.

**Uniqueness.** On the `liquidus` branch, raising $T$ melts ice (which takes $L(T)$ per kilogram) and warms everything, so $H(T)$ rises strictly; on the `liquid` branch it rises with the heat capacity alone. $H$ is continuous where the branches meet, so $H(T)$ is strictly increasing: **for each $H$ there is exactly one temperature**, found by bracketing on $(T_E, 200\,^\circ\text{C})$. (7) is evaluated at both ends of the eutectic plateau with the same $w_E$ — computed from (11), not typed — or the branch and the plateau fail to meet by $10^{-3}$ kJ and the bracket breaks.

## 5.3 The latent heat that the bath actually charges

On the liquidus with $S$ fixed, $m_l = S(1-w)/w$, so melting $\mathrm{d}m$ of ice changes the composition by $\mathrm{d}w/\mathrm{d}m = -w^2/S$, and the temperature follows the liquidus, $\mathrm{d}T_b = \mathrm{d}w/w_L'(T_b)$. The heat taken per kilogram melted is the derivative of (7):

$$\frac{\mathrm{d}T_b}{\mathrm{d}m_{\text{melt}}} = \frac{w^2}{S\,|w_L'(T_b)|}, \qquad L_{\text{eff}} \equiv \frac{\mathrm{d}H}{\mathrm{d}m_{\text{melt}}} = \underbrace{L_0 + (c_b - c_i)\,T_b}_{L(T_b)} + \underbrace{C_{\text{bath}}\,\frac{\mathrm{d}T_b}{\mathrm{d}m_{\text{melt}}}}_{\text{warming the bath}}. \tag{13}$$

*Where it comes from.* The first term is Kirchhoff's latent heat (6) at the bath temperature, with the meltwater joining brine of heat capacity $c_b$; the second is the heat spent warming the whole bath, of heat capacity $C_{\text{bath}}$, along the liquidus while that kilogram melts (the dilution raises the freezing point). (7) contains both automatically; the code evaluates $L_{\text{eff}}$ by finite difference in $H$. Both factors in $\mathrm{d}T_b/\mathrm{d}m$ carry $1/S$: **double every mass and the drift per kilogram melted halves**, the $1/\text{size}$ law of the rig note.

## 5.4 Where the cold comes from

Integrating (7) between the unmixed ingredients and the mixed bath at constant $H$, the heat that cools the bath is paid by only two terms: $L(T)\,m_{\text{melt}}$, the ice melting, and $s\,\Delta h_s$, the salt dissolving. Their ratio can be read off any equilibrium; for the charges in the earlier notes the salt supplies about a tenth of the cold and the ice the rest (§12).

## 5.5 Reaching the eutectic is an energy condition

Adiabatic mixing conserves $H$. A charge reaches the thermostat only if its starting enthalpy lies below the top of the eutectic plateau, and keeps solid salt only if it lies further below:

$$H_0 \;\le\; H(T_E;\ m_l = \min(W,\ S/r_E)), \tag{14}$$

with $H_0$ the enthalpy of the ingredients at their own starting temperatures.

*Where it comes from.* The plateau's top is the state with all the water that can be brined at $w_E$ already liquid; any more enthalpy and the bath leaves the plateau. Free water is expensive: every kilogram of it must be cooled to $-21.2\,^{\circ}\mathrm{C}$ and salted to $w_E$, and the only payment is melting ice, which brings more water. Checking that the salt can saturate the free water — a mass condition — is necessary but not sufficient.

# 6 How heat moves

## 6.1 Conduction

In a solid or a still fluid heat flows down the temperature gradient. Fourier's law,

$$\mathbf{q} = -k\,\nabla T, \tag{15}$$

gives the heat flux $\mathbf{q}$ ($\mathrm{W\,m^{-2}}$), with $k$ the thermal conductivity ($\mathrm{W\,m^{-1}K^{-1}}$). Energy conservation (3) for a small volume, whose enthalpy changes at $\rho c\,\partial T/\partial t$ per unit volume, turns it into the heat equation

$$\rho c\,\frac{\partial T}{\partial t} = \nabla\cdot(k\nabla T), \qquad \frac{\partial T}{\partial t} = \alpha\,\nabla^2 T, \qquad \alpha = \frac{k}{\rho c}. \tag{16}$$

*Where it comes from.* The net heat flowing into a small volume through its faces is $-\nabla\cdot\mathbf{q}$ per unit volume; it raises the temperature of the mass $\rho$ per unit volume of heat capacity $c$. The thermal diffusivity $\alpha$ ($\mathrm{m^2\,s^{-1}}$) is the only combination of the three properties that appears; in time $t$ a change of temperature diffuses a distance

$$\delta(t) \sim \sqrt{\alpha t}, \tag{17}$$

because $\partial/\partial t \sim 1/t$ and $\nabla^2 \sim 1/\delta^2$ must balance (the exact solution for a suddenly heated wall is an error function of $x/2\sqrt{\alpha t}$). For eutectic brine $k_b = 0.54\,\mathrm{W\,m^{-1}K^{-1}}$, $\rho_b = 1182\,\mathrm{kg\,m^{-3}}$, $c_b = 3307\,\mathrm{J\,kg^{-1}K^{-1}}$, so $\alpha_b = 0.14\,\mathrm{mm^2\,s^{-1}}$ and in three minutes conduction alone carries heat **5 mm**; for ice $\alpha = 1.2\,\mathrm{mm^2\,s^{-1}}$.

## 6.2 Steady conduction through layers

When nothing stores heat, the same heat flow $\dot{Q}$ passes through every layer in a stack. Integrating (15) across a flat slab of thickness $\delta$ and area $A$, and across a cylindrical shell of length $\lambda$ between radii $r_1$ and $r_2$ (where the flux falls as $1/r$ because the same $\dot{Q}$ crosses every circle $2\pi r\lambda$):

$$\dot{Q} = \frac{kA}{\delta}\,\Delta T, \qquad \dot{Q} = \frac{2\pi k\lambda}{\ln(r_2/r_1)}\,\Delta T. \tag{18}$$

The coefficient of $\Delta T$ is a conductance $G$ ($\mathrm{W\,K^{-1}}$); its inverse is a resistance. Since the temperature drops of layers in series add while the heat flow is common, **resistances in series add**; side by side, the heat flows add, so **conductances in parallel add**. Every heat path in the rig is built from these two rules.

## 6.3 Convection and the film coefficient

At a wall in a moving fluid, the fluid touching the wall is at rest and passes heat only by conduction, through a thin *thermal boundary layer* of thickness $\delta_T$ in which the temperature changes from the wall's $T_w$ to the fluid's $T_\infty$. Outside it the flow mixes the fluid. So

$$q = h\,(T_w - T_\infty), \qquad h \approx \frac{k}{\delta_T}, \qquad \mathrm{Nu} = \frac{h\mathcal{L}}{k} \approx \frac{\mathcal{L}}{\delta_T}. \tag{19}$$

*Where it comes from.* The first equation, Newton's law of cooling, *defines* the film coefficient $h$ ($\mathrm{W\,m^{-2}K^{-1}}$); the physics is in how $\delta_T$ depends on the flow, and the Nusselt number Nu measures it against a length $\mathcal{L}$ of the body.

## 6.4 Free convection

Nothing stirs the bath or the drink: fluid moves only because warm fluid is lighter. Next to a wall at $T_w$ the fluid's density differs by $\Delta\rho = \rho\beta\,\Delta T$ ($\beta$ the thermal expansion coefficient), so the boundary layer feels a buoyancy force $\rho g\beta\Delta T$ per unit volume. In a viscous liquid ($\mathrm{Pr} = \nu/\alpha \gg 1$: 7–13 for the drink, 33 for cold brine) friction balances it, $\rho g\beta\Delta T \sim \mu u/\delta_T^2$, which sets the speed $u$ of the rising (or sinking) layer. Heat carried along the wall over a length $\mathcal{L}$ balances heat conducted across the layer, $u\,\Delta T/\mathcal{L} \sim \alpha\,\Delta T/\delta_T^2$. Eliminating $u$:

$$\delta_T \sim \mathcal{L}\,\mathrm{Ra}^{-1/4}, \qquad \mathrm{Nu} \sim \mathrm{Ra}^{1/4}, \qquad \mathrm{Ra} = \frac{g\beta\,\Delta T\,\mathcal{L}^3}{\nu\alpha}, \qquad \Longrightarrow \qquad h \propto \left(\frac{\Delta T}{\mathcal{L}}\right)^{1/4}. \tag{20}$$

*Where it comes from.* Two balances, buoyancy against friction and convection against conduction, give the thickness of the layer; the Rayleigh number Ra is the ratio of the two. Two consequences are used directly. **(i)** The film coefficient grows as $\Delta T^{1/4}$: the drink-side film in the simulator follows $h = h_{\mathrm{ref}}(\Delta T/\Delta T_0)^{1/4}$ with $\Delta T_0 = 25\,\mathrm{K}$. **(ii)** It shrinks as $\mathcal{L}^{-1/4}$: a can lying on its side, for which $\mathcal{L}$ is the 6.6 cm diameter rather than the 11.7 cm height, has films $(11.66/6.6)^{1/4} = 1.15$ times stronger from the scaling alone.

*When it holds.* For laminar boundary layers, $\mathrm{Ra} \lesssim 10^9$: here Ra is $10^7$–$10^8$.

The scaling fixes the exponents; the constants come from experiment. Churchill & Chu (1975) fitted, for a vertical wall of height $\mathcal{L}$ and a horizontal cylinder of diameter $D$,

$$\mathrm{Nu}_{\mathcal{L}} = 0.68 + \frac{0.670\,\mathrm{Ra}^{1/4}}{[1+(0.492/\mathrm{Pr})^{9/16}]^{4/9}}, \qquad \mathrm{Nu}_D = \left\{0.60 + \frac{0.387\,\mathrm{Ra}_D^{1/6}}{[1+(0.559/\mathrm{Pr})^{9/16}]^{8/27}}\right\}^2.$$

For brine at $-15\,^\circ$C on Can 2 these give $h = 240$–$358\,\mathrm{W\,m^{-2}K^{-1}}$ upright and 280–457 lying down, for a 5–25 K difference: a film 1.5–2.2 mm thick, and a ratio of 1.17–1.28 between the two orientations (1.15–1.33 for the drink inside). The simulator takes the brine-side $h_o = 270\,\mathrm{W\,m^{-2}K^{-1}}$ and the drink-side $h_i = 400$ for an upright can (a bottle study, §17), and multiplies both by `on_side_factor` $= 1.2$ for a can on its side.

## 6.5 Radiation

Two facing grey surfaces at absolute temperatures $T_1$ and $T_2$ exchange $q = \varepsilon_{\mathrm{eff}}\sigma(T_1^4 - T_2^4)$, with $\sigma = 5.67\times10^{-8}\,\mathrm{W\,m^{-2}K^{-4}}$ and $1/\varepsilon_{\mathrm{eff}} = 1/\varepsilon_1 + 1/\varepsilon_2 - 1$. Factoring $T_1^4 - T_2^4 = (T_1 - T_2)(T_1 + T_2)(T_1^2 + T_2^2) \approx 4\bar{T}^3(T_1 - T_2)$ for close temperatures,

$$q \approx h_{\mathrm{rad}}\,(T_1 - T_2), \qquad h_{\mathrm{rad}} = 4\,\varepsilon_{\mathrm{eff}}\,\sigma\,\bar{T}^3, \tag{21}$$

5.0 $\mathrm{W\,m^{-2}K^{-1}}$ at 290 K for $\varepsilon = 0.9$ and 3.7 at 270 K between two such surfaces. Radiation therefore adds to convection like a parallel film. The box's air films (§10) are built this way: room air on the foam, $h \approx 5 + 4 = 9$; the 1 cm gap between foam and glass, conduction $k_{\mathrm{air}}/\delta = 0.024/0.01 = 2.4$ plus radiation 3.5, $h \approx 6$; the still air under the lid, warm above cold so that it cannot convect, $h \approx 4$.

## 6.6 The drink as one temperature

The drink is lumped: one temperature $T_d$ for the whole can. By conduction alone it would not be uniform — the Biot number $\mathrm{Bi} = h_i\,(V/A)/k_w = 400 \times 0.0129/0.58 = 8.9$ compares the film outside

with conduction inside and is well above 0.1 — but free convection inside the can (Ra $\sim 10^8$) mixes it; the lumped temperature is the volume average of the full flow problem solved in COMSOL. Energy conservation (3) for the drink and can, of total heat capacity $C_d = m_d c_d + m_{\text{can}} c_{\text{Al}} = 1332\,\text{J K}^{-1}$, losing heat through the can wall of area $A$:

$$C_d \frac{\mathrm{d}T_d}{\mathrm{d}t} = -UA\,(T_d - T_b), \qquad \frac{1}{U} = \frac{1}{h_i} + \frac{\delta_{\text{Al}}}{k_{\text{Al}}} + \frac{1}{h_o}. \tag{22}$$

*Where it comes from.* The heat flows from the drink, across the inner film, the aluminium and the outer film in series (§6.2); the wall's resistance, $10^{-4}/237 = 4 \times 10^{-7}\,\text{m}^2\text{K W}^{-1}$, is negligible beside $1/h \approx 3 \times 10^{-3}$. For a constant bath temperature and constant $U$, (22) is solved by an exponential, and with $h \propto \Delta T^{1/4}$ on both sides ($U = U_0\,(\theta/\Delta T_0)^{1/4}$, $\theta = T_d - T_b$) by separation of variables, $\theta^{-5/4}\mathrm{d}\theta = -\mathrm{d}t/(\tau \Delta T_0^{1/4})$:

$$\tau = \frac{C_d}{UA}, \qquad t_{\text{target}} = \tau \ln \frac{T_{d0} - T_b}{T_{\text{target}} - T_b}, \qquad \theta(t) = \theta_0 \Big(1 + \frac{t}{t^*}\Big)^{-4}, \quad t^* = 4\tau \Big(\frac{\Delta T_0}{\theta_0}\Big)^{1/4}. \tag{23}$$

With $U_0 = 160\,\text{W m}^{-2}\text{K}^{-1}$ and $A = 310\,\text{cm}^2$, $\tau = 268\,\text{s}$. Both closed forms are used to check the integrators, not as the model.

# 7 Melting, dissolving and freezing: the rates

## 7.1 The Stefan condition at the ice surface

Ice melts at its surface, and the surface can store no heat. Applying (3) to a thin layer straddling it: heat arrives from the brine across the brine-side film, $h_{bi} A_i (T_b - T_s)$, and from the ice's own interior, $G_c (T_{\text{ice}} - T_s)$; it is all spent turning ice at $T_s$ into water at $T_s$:

$$\dot{m}_{\text{melt}}\,[h_w(T_s) - h_i(T_s)] = h_{bi} A_i\,(T_b - T_s) + G_c\,(T_{\text{ice}} - T_s). \tag{24}$$

*Where it comes from.* Energy conservation at a moving phase boundary (the Stefan condition). $A_i$ is the wetted ice area, $h_{bi} = 100\,\text{W m}^{-2}\text{K}^{-1}$ the brine-side film coefficient for the slow free convection among the pieces, $T_s$ the surface temperature, $T_{\text{ice}}$ the interior temperature of the pieces and $G_c$ the conductance from interior to surface. A negative right-hand side freezes brine onto the ice. The interior cools as it gives heat to the surface: $C_{\text{ice}}\,\mathrm{d}T_{\text{ice}}/\mathrm{d}t = -G_c(T_{\text{ice}} - T_s)$.

**Area and interior conductance.** The heap is $n$ cubes of side $a$, so $a = (m_i/n\rho_i)^{1/3}$ and the wetted area is $A_i = 6na^2\,\phi_{\text{wet}}$, with $\phi_{\text{wet}}$ the fraction of the heap below the liquid surface (§9). As the pieces shrink, $A_i \propto m_i^{2/3}$. Heat inside a piece crosses a mean distance comparable with its volume-to-area ratio $a/6$, so $G_c = k_i A_i/(a/6)$ (for a sphere the exact quasi-steady value is a fifth of the radius, $0.6$ times this distance, so $G_c$ is if anything an underestimate). The interior's time constant $C_{\text{ice}}/G_c$ is about 9 s for 2 cm pieces.

## 7.2 The temperature of the ice surface

**Bulk** (default): the surface sits at the freezing point of the brine around it, $T_s = T_f(w)$ from (11). This assumes salt reaches the surface as fast as melting dilutes it there.

**Salt-limited**: meltwater is fresh, so it dilutes the brine at the surface to $w_s < w$, whose freezing point is higher. Three conditions fix $T_s$, $w_s$ and the melt rate: the Stefan condition (24); the liquidus at the

surface, $T_s = T_f(w_s)$; and a salt balance at the surface,

$$\rho\,\gamma_S A_i\,(w - w_s) = \dot{m}_{\text{melt}}\,w_s, \qquad \gamma_S = \frac{\gamma_T}{\mathcal{R}}, \quad \gamma_T = \frac{h_{bi}}{\rho c_b}, \tag{25}$$

*Where it comes from.* Salt diffuses to the surface across the boundary layer with a transfer velocity $\gamma_S$ (m s$^{-1}$), the mass-transfer analogue of $h/\rho c$, and must arrive fast enough to bring the meltwater, $\dot{m}$ per second, up to $w_s$. This is the three-equation ice–ocean interface of Holland & Jenkins (1999). Salt diffuses about a hundred times more slowly than heat ($\alpha/D_{\text{salt}} \approx 0.14\times10^{-6}/1.5\times10^{-9}$), and across a boundary layer the transfer velocities scale with the diffusivities to the power $2/3$, so $\mathcal{R} = \gamma_T/\gamma_S \approx 100^{2/3} \approx 20$; the literature gives 10–200, the default is 30.

### 7.3 Salt dissolving

A salt grain dissolves as salt diffuses away from its surface, where the brine is saturated, into the bulk brine. Fick's law, the diffusion analogue of (15), across a concentration boundary layer of thickness $\delta_c$, gives a mass flux $j = D\rho_b(w_{\text{sat}} - w)/\delta_c$, so

$$\dot{s} = k_d\,A_s\,\rho_b\,(w_{\text{sat}}(T) - w), \qquad k_d = \frac{D}{\delta_c}, \qquad A_s = A_{s0}\Big(\frac{S-s}{S}\Big)^{2/3}. \tag{26}$$

*Where it comes from.* $k_d$ (m s$^{-1}$) is the mass-transfer coefficient: with $D = 1.5\times10^{-9}$ m$^2$ s$^{-1}$ and $\delta_c \approx 0.1$ mm in stirred brine, $k_d \approx 1.5\times10^{-5}$ (default $10^{-5}$); an unstirred layer ten times thicker gives $10^{-6}$. The grains are cubes of side 0.5 mm, whose total area $A_{s0} = 6S/(\rho_s a_s) = 5.5\,\text{m}^2$ for 1 kg shrinks as their mass to the $2/3$. Above saturation the grains grow back. While there is ice, dissolving stops at $w_E$: beyond it salt would only come out again as hydrohalite (§4.3). Each kilogram that dissolves takes $\Delta h_s$ from the brine, as (7) records.

### 7.4 The drink freezing

The drink is mostly water with sugar ($\approx 10$ %) and dissolved CO$_2$. By (10) with $i = 1$, 10 % sucrose ($b = 0.33$ mol kg$^{-1}$) lowers its freezing point by 0.6 K; the CO$_2$ adds about 0.3 K: $T_{f0} = -0.9\,^\circ$C. When a fraction $\varphi$ of the drink's water has frozen, the solutes are all in the remaining liquid, so their molality is $1/(1-\varphi)$ times larger and, by the same law,

$$T_f(\varphi) = \frac{T_{f0}}{1-\varphi}, \qquad H_d = C_d T - \varphi\,m_{dw}\,L(T) \quad \text{on} \quad T = T_f(\varphi). \tag{27}$$

*Where it comes from.* Freezing concentrates the solution, so the drink freezes over a range of temperatures, not at one: 50 % of its water is frozen at $-1.8\,^\circ$C, 90 % at $-9\,^\circ$C. Its enthalpy is that of the liquid less the latent heat (6) of the ice formed, with $m_{dw} = 0.89\,m_d$ the drink's water; given $H_d$ from (3), the model solves for $T$ and $\varphi$ together. A still drink can supercool: it stays liquid down to $T_{f0} - \Delta T_{\text{sc}}$, and when ice nucleates the latent heat released warms it back onto the curve.

## 8 Where the drink's heat goes: spreading through the bath

### 8.1 Why the heat cannot reach the whole bath at once

Once Can 2 is in, about 217 W leaves it ($T_d = 26.4\,^\circ$C, with the brine next to it already at $-17.9\,^\circ$C). A perfectly mixed bath would share that at once among 2.55 kg of ice and 3.97 kg of brine. But by (17) conduction carries heat only 5 mm through still brine in the three minutes the drink takes to cool. The brine next to the can warms, its ice melts first, and the drink faces warmer brine than the bulk.

## 8.2 How far the heat gets before the ice takes it

Brine warmer than the ice surface loses heat to the ice at $h_{bi}a_v(T - T_s)$ per unit volume, with $a_v$ the wetted ice area per unit volume of bath (135 m$^{-1}$ at drink-in). In a steady state near the can, heat conducted outwards at an effective conductivity $k_{\text{eff}}$ is absorbed by the ice on the way. Energy conservation (16) with this sink, in one dimension $x$ out from the can:

$$k_{\text{eff}}\,\frac{\mathrm{d}^2T}{\mathrm{d}x^2} = h_{bi}a_v\,(T - T_s) \implies T - T_s \propto e^{-x/\ell}, \qquad \ell = \sqrt{\frac{k_{\text{eff}}}{h_{bi}\,a_v}}. \tag{28}$$

*Where it comes from.* Conduction in, absorption by melting out: the temperature excess decays exponentially over the *penetration length* $\ell$, 0.6 cm for still brine and 2.8 cm for $k_{\text{eff}} = 20\,k_b$. Most of the drink's heat is absorbed within a few centimetres of the can.

## 8.3 The ice bed conducts and convects

Still brine is only a lower bound for $k_{\text{eff}}$. Two things raise it. First, ice conducts four times better than brine. A bed of porosity $\varepsilon = 0.4$ conducts between the series and parallel limits of its two phases,

$$\left(\frac{\varepsilon}{k_b} + \frac{1-\varepsilon}{k_i}\right)^{-1} \le k_{\text{bed}} \le \varepsilon k_b + (1-\varepsilon)k_i, \tag{29}$$

0.98–1.54 W m$^{-1}$K$^{-1}$, or 2–3 times $k_b$: the two limits are layers of brine and ice across and along the heat flow, in series and in parallel (§6.2). Second, the brine moves through the bed. Flow through a porous bed obeys Darcy's law, speed proportional to the driving pressure gradient, here buoyancy: $u \sim Kg\beta\Delta T/\nu$, with $K$ the permeability (m$^2$). Modelling the gaps as channels of width $\propto \varepsilon a/(1-\varepsilon)$ with Poiseuille flow gives the Kozeny–Carman form $K = a^2\varepsilon^3/180(1-\varepsilon)^2 = 4\times10^{-7}$ m$^2$ (the 180 is empirical). Repeating the boundary-layer balance of §6.4 with Darcy's law in place of viscous friction, $u\,\Delta T/\mathcal{L} \sim \alpha_m \Delta T/\delta^2$ with $u$ now independent of $\delta$, gives

$$\mathrm{Nu} \sim \mathrm{Ra}_D^{1/2}, \qquad \mathrm{Ra}_D = \frac{g\beta\,\Delta T\,K\,\mathcal{L}}{\nu\,\alpha_m}, \tag{30}$$

with mean $\mathrm{Nu} = 0.89\,\mathrm{Ra}_D^{1/2}$ for a heated vertical wall (Cheng & Minkowycz 1977) and $\alpha_m$ the bed's diffusivity. For a 14.5 cm bed and $\Delta T = 1$–6 K, $\mathrm{Ra}_D \approx 10^2$–$10^3$ and Nu $\approx$ 9–27. Together, the *convection factor* $f = k_{\text{eff}}/k_b \approx (k_{\text{bed}}/k_b)\,\mathrm{Nu} \approx 20$–60. The estimate is rough — the gaps (about 1 cm) are not small next to the 2 mm film on the can, and warm brine rising past the can spreads along the top of the bath — so $f$ is an input; the default 20 is the low end. $f = 1$ is still brine, $f = \infty$ a perfectly mixed bath.

## 8.4 Rings round the can

The bath is divided into $N$ rings (default 8): shells round the can's axis from its wall, $r_c = 3.30$ cm, out to $R = \sqrt{A_f/\pi} = 11.45$ cm, the radius of a circle with the floor area $A_f$ of the tank. Each ring is a finite volume obeying (3), with its own brine temperature $T_j$, ice, ice-interior temperature and crystals; the salt fraction $w$ is shared, since the brine that carries heat also mixes salt. The edges are spaced evenly in $\ln r$,

$$r_j = r_c\,(R/r_c)^{j/N}, \qquad j = 0, \dots, N, \tag{31}$$

so the rings are thin by the can (0.56 cm) and wide at the walls (1.65 cm), where little happens. Ring $j$ holds the share $\psi_j = (r_{j+1}^2 - r_j^2)/(R^2 - r_c^2)$ of the brine, ice and crystals. The drink heats only the

inner ring; the glass, the lid, the gaps in the box and the stirrer heat only the outer one:

$$C_j\,\dot{T}_j = G\,(T_{j-1}-T_j) + G\,(T_{j+1}-T_j) - Q_{i,j} - \Delta h_s\,\dot{s}_j \\ + \delta_{j0}\,UA\,(T_d - T_0) + \delta_{j,N-1}\,(Q_{\text{glass}} + Q_{\text{lid}} + Q_{\text{gaps}} + P_{\text{stir}}), \tag{32}$$

$$G = \frac{2\pi\,k_{\text{eff}}\,\lambda}{\ln(r_{j+1}/r_j)} = \frac{2\pi N k_{\text{eff}}\,\lambda}{\ln(R/r_c)}, \qquad k_{\text{eff}} = f\,k_b(w). \tag{33}$$

*Where it comes from.* (32) is (3) for ring $j$, of heat capacity $C_j$: heat conducted in from each neighbour, heat lost to its ice ($Q_{i,j}$, (24)) and to dissolving its crystals, and the drink or the room where they touch it ($\delta$ is 1 for the named ring and 0 otherwise). $G$ is the cylindrical-shell conductance (18) between the middles of neighbouring rings, which are one step of $\ln r$ apart; with even spacing in $\ln r$ it is the same for every pair. $\lambda$ is the length of the shells: the can's own length (11.66 cm) for a can on its side, the depth of brine for an upright one. At drink-in, $G = 51\,\mathrm{W\,K^{-1}}$; from can to walls the bath adds $\ln(R/r_c)/(2\pi k_{\text{eff}}\lambda) = 0.16\,\mathrm{K\,W^{-1}}$ in series with the can's own $1/UA = 0.20\,\mathrm{K\,W^{-1}}$, although by (28) most of the heat is absorbed before the walls. The rings must resolve $\ell$: 8 rings are within 4 % of 50 at the default, but still brine needs 50 (§13).

# 9 The liquid level and the areas that follow it

Every heat path into or out of the brine crosses a wetted surface, and most of those surfaces depend on how high the liquid stands. At every step the level is found from the volumes, and each area from the level.

## 9.1 Volumes

The liquid occupies $V_l = (m_l + s)/\rho_b(w) + (S - s)/\rho_s$: brine of density $\rho_b = 1000 + 786\,w\,\mathrm{kg\,m^{-3}}$ (Melinder), and the crystals on the floor, $\rho_s = 2165$. Dissolving a kilogram of salt changes the brine volume by $[\rho_b - 786(1-w)]/\rho_b^2 = 0.41\,\mathrm{L}$, less than the 0.46 L of the crystal, and makes the brine denser.

## 9.2 A heap resting on the floor, or floating

The crushed ice is a heap of porosity $\varepsilon$: a fraction $1-\varepsilon$ of its volume is ice. Its thickness is $t_h = m_i/[\rho_i(1-\varepsilon)A_f]$, 17.7 cm for 4 kg in a 412 $\mathrm{cm^2}$ tank. While it rests on the floor the liquid has only the gaps; once there is enough of it, the heap floats. By Archimedes, a floating heap displaces its own weight of brine, $V_{\text{sub}} = m_i/\rho_b$, wherever its ice is. The level $Z$ is where the volume below it, $A_f Z$, holds the liquid and the submerged ice:

$$Z = \frac{V_l}{\varepsilon A_f}\ \text{(resting)}, \qquad Z = \frac{V_l + m_i/\rho_b}{A_f}\ \text{(floating)}, \tag{34}$$

and the part of the can under the surface adds its own volume over $A_f$.

*Where it comes from.* Volume conservation and Archimedes. The heap lifts off when the liquid fills its gaps to the depth at which it would float, $V_l = \varepsilon A_f\,m_i/[\rho_b(1-\varepsilon)A_f]$; at that point both expressions give the same $Z$, so the level is continuous. The wetted fraction of the ice is $Z/t_h$ while resting and $\rho_i/\rho_b$ while floating, again continuous.

With the defaults the level rises from 12.5 to 13.7 cm in the first half-minute: the heap still rests on the floor, and water melting off the ice fills its gaps, raising the level 2.5 times faster per litre than open liquid would. The brine then becomes dense enough to float the heap, and as the rest of the

salt dissolves, the denser brine lifts the floating ice, so the level eases back to 13.4 cm. That is Archimedes for a bath mixed all at once at $t = 0$; a real bath, made up over a minute, rises over that minute. As floating ice melts later, its meltwater takes a little more room than the brine it displaced, and the level creeps up about 2 mm.

### 9.3 The can in the bath

In the rig the can lies on its side, along the tank. The part of its circular end below a depth $y$ measured from its lowest point is a circular segment, a sector of the circle less the triangle over the chord:

$$S(y) = r_c^2 \arccos\!\left(\frac{r_c - y}{r_c}\right) - (r_c - y)\sqrt{2r_c y - y^2}, \qquad V_{\text{under}}(y) = H_c\, S(y), \tag{35}$$

with $H_c = 11.66$ cm the can's length ($V_{\text{under}} = \pi r_c^2 y$ for an upright can). With the can's lowest point at $z_c$, the depth under the surface $y$ satisfies $y = Z_0 - z_c + V_{\text{under}}(y)/A_f$, because the displaced volume raises the level ($Z_0$ is the level without the can); this has one root, since $H_c$ times the chord is less than $A_f$. A can held on its holder comes to rest at $z_c = 1$ cm; a free can floats where $\rho_b V_{\text{under}} = m_{\text{can}}$ (68 % of its height under, lying down). It is pushed down from the surface over 10 s, so its 0.40 L raises the level by $0.40/412 = 0.97$ cm smoothly, within the first 5 s: lying down it is only 6.6 cm high.

### 9.4 The can's heat-passing area

Heat crosses the can wall only where there is liquid on both sides: brine outside, up to the brine's surface, and drink inside, up to the drink's own surface. Above the drink is gas, a headspace of $h_s = 1.2$ cm measured upright, which passes little heat. Lying down, the same 41 cm$^3$ of gas makes a strip of depth $h_g$ along the top, found from $H_c\, S(h_g) = \pi r_c^2 h_s$: $h_g = 1.05$ cm. The can then passes heat through both ends and the curved wall below the strip:

$$A_c = 2\, S(y) + 2r_c \arccos\!\left(\frac{r_c - y}{r_c}\right) H_c, \qquad y = \min(d,\ 2r_c - h_g), \tag{36}$$

with $d$ the depth of can under the brine.

*Where it comes from.* The ends contribute two segments of depth $y$ and the curved wall an arc of length $2r_c \arccos[(r_c - y)/r_c]$ along the whole length. Submerged, Can 2 passes heat through 240 cm$^2$ of its 310 cm$^2$. Standing upright it would be the bottom and the side up to the drink, $\pi r_c^2 + 2\pi r_c \min(d, H_c - h_s) = 251$ cm$^2$. Either way the area grows smoothly with the level.

**Glass.** The brine wets the floor and the walls up to $Z$: $A_g = A_f + 2(W_t + \ell_t)Z$, with $W_t$ and $\ell_t$ the tank's width and length; it passes heat across the brine film and half the glass thickness, $1/U_g = 1/h_{bg} + \delta_g/2k_g$, with $h_{bg} = 270$ W m$^{-2}$K$^{-1}$.

## 10 The box: heat from the room

The room is 48 K warmer than the bath, so heat leaks in through the styrofoam all the time.

### 10.1 A layer of foam that stores heat

Each path through the box is a series of layers (§6.2): the room's air film, the foam, and whatever is inside. The foam also holds heat, $C_f = \rho_f c_f A t$ (280 J K$^{-1}$ for the whole box). Lumping that heat at the middle of the foam splits the path in two:

$$G_{\text{out}} = \frac{A}{1/h_{\text{out}} + t/2k_f}, \quad G_{\text{in}} = \frac{A}{t/2k_f + 1/h_{\text{in}}}, \quad C_f \frac{\mathrm{d}T_f}{\mathrm{d}t} = G_{\text{out}}(T_{\text{room}} - T_f) - G_{\text{in}}(T_f - T_{\text{inside}}). \tag{37}$$

*Where it comes from.* (3) for the foam node, with each half of the foam ($t/2$ of conductivity $k_f = 0.035\,\mathrm{W\,m^{-1}K^{-1}}$) in series with its film. In steady state the node passes on what it receives and the two halves add back to the full series resistance, $UA = A/(1/h_{\mathrm{out}} + t/k_f + 1/h_{\mathrm{in}})$. Areas are taken at the middle of the foam.

## 10.2 Four paths

| Path | Leak |
|---|---|
| foam walls → air gap → tank glass → brine | $0.20\,\mathrm{W\,K^{-1}}$ |
| foam floor → tank glass → brine (the tank stands on the foam: no air gap) | $0.07\,\mathrm{W\,K^{-1}}$ |
| foam lid → still air → brine | $0.05\,\mathrm{W\,K^{-1}}$ |
| gaps → brine: lid seam, the lid's thin lip, holes for probe leads | $0.02\,\mathrm{W\,K^{-1}}$ |
| all of them | $0.34\,\mathrm{W\,K^{-1}}$ |

The gaps bypass the foam. Air in the box is cold and dense, so through a slot of height $\delta_s$ and width $W_s$ in the lid seam it flows out at the bottom of the slot and room air flows in at the top. The pressure difference driving it grows linearly with height from the middle of the slot, $\Delta p = \Delta\rho\, g\, z$, and Bernoulli's equation turns it into a speed $u = \sqrt{2\Delta p/\rho} = \sqrt{2g'z}$, with $g' = g\Delta T/T$; integrating over the half-height that flows each way,

$$Q_{\mathrm{air}} = \frac{C_D}{3}\, W_s\, \delta_s \sqrt{g'\delta_s}, \tag{38}$$

*Where it comes from.* Hydrostatics and Bernoulli, with a discharge coefficient $C_D \approx 0.6$ for the contraction of the jet (Linden 1999). For a 1 mm seam round the lid ($W_s = 1.1$ m) it carries $\rho c_p Q_{\mathrm{air}} \Delta T \approx 0.2$ W, 0.005–0.01 $\mathrm{W\,K^{-1}}$ of leak; a 5 mm gap carries $5^{3/2} = 11$ times as much. With the lid's thin lip ($\approx 0.01\,\mathrm{W\,K^{-1}}$) the gaps are estimated at $0.02\,\mathrm{W\,K^{-1}}$.

## 10.3 Measuring the leak: the box test

Fill the tank with cold water only, close the box, and log the water. After the first half-hour, while the room-temperature glass gives up its heat, the water and glass (heat capacity $C$) warm only by the leak, $C\,\mathrm{d}T/\mathrm{d}t = (UA)_{\mathrm{box}}(T_{\mathrm{room}} - T)$, whose solution approaches the room exponentially. Taking logarithms at two times,

$$(UA)_{\mathrm{box}} = \frac{C}{t_2 - t_1}\,\ln\frac{T_{\mathrm{room}} - T(t_1)}{T_{\mathrm{room}} - T(t_2)}, \qquad C = m_w c_w + C_{\mathrm{glass}} = m_w \times 4190 + 2020\ \mathrm{J\,K^{-1}}. \tag{39}$$

The model predicts 6.7 °C at 30 min and 9.4 °C at 3 h for 5 kg of water from 4 °C, and (39) applied to its own curve gives back its leak to 1 %. `box_test.toml` sets the run up; a measured value goes into `[box] measured_UA`, which scales every path to add up to it.

# 11 Putting it together: two models and how they are solved

## 11.1 The equilibrium model, `b8_sim.py`

The bath is one body at equilibrium, described by its enthalpy $H_b$; the drink is a lump. Energy conservation (3) for each, with the drink's loss (22) as the bath's gain:

$$\begin{aligned} C_d\,\dot{T}_d &= -\,U A\,(T_d - T_b), \\ \dot{H}_b &= \underbrace{U A\,(T_d - T_b)}_{\text{from the drink}} + \underbrace{(UA)_{\text{box}}\,(T_{\text{room}} - T_b)}_{\text{leak}} + \underbrace{P_{\text{stir}}}_{\text{stirrer}}, \qquad T_b = \mathcal{T}(H_b), \end{aligned} \tag{40}$$

with $U = U_0$ (Newton) or $U = U_0\,(|T_d - T_b|/\Delta T_0)^{1/4}$ (free convection). Salt added during a run enters as $\Delta S$ with $\Delta H = \Delta S\, c_s T_{\text{room}}$. The pair is integrated by Heun's method (a predictor step and a trapezoidal corrector, second-order accurate) at $\Delta t = 0.25$ s; against the closed forms (23) it reproduces the time to 5 °C to 0.01 s.

## 11.2 The time-dependent model, `b8_model.py`

The simulator follows, as functions of time: the drink's enthalpy (and whether ice has nucleated in it); the liquid water and dissolved salt; for each ring, its brine enthalpy, its crystals, its ice and the ice's interior temperature; the glass temperature; and the temperatures of the foam walls and lid. Each obeys (3) or a mass balance, with the flows of §6–10: the drink (22) and (27); the rings (32); the ice (24); the crystals (26); the glass $C_g\dot{T}_g = Q_{\text{walls}} - U_g A_g (T_g - T_b)$; the foam (37). After each melt or dissolution the brine is shared out again between the rings in proportion to their volume, carrying its heat with it.

**Time stepping.** Each quantity is stepped forward by Euler's method, $y_{n+1} = y_n + \Delta t\, \dot{y}(y_n)$, with $\Delta t = 0.5$ s. Euler's method is stable for a process of time constant $\tau$ only if $\Delta t < 2\tau$, and accurate if $\Delta t \ll \tau$. The fastest processes are the interior of the ice pieces (9 s), the brine next to the can (60 s) and the glass (75 s); the program checks these at the start and warns if $\Delta t$ is too long. The spreading between rings can be much faster when the inner ring is thin, so it alone is stepped backward,

$$C_j(T_j^{n+1} - T_j^{*}) = G\,\Delta t\,(T_{j-1}^{n+1} - 2T_j^{n+1} + T_{j+1}^{n+1}),$$

a tridiagonal system solved exactly in one sweep: stable for any $G$, and since each flux appears with both signs, it conserves energy exactly. With $f = \infty$ all rings take one temperature, which reproduces the first version of the simulation (`inputs_original.toml`) to $10^{-13}$ K.

**Energy bookkeeping.** After every run the change in total enthalpy is compared with the heat let in from the room and the stirrer (3). It closes to $10^{-9}$ J of the 43 kJ let in. That proves the bookkeeping, not the physics: a wrong heat-transfer coefficient conserves energy just as well. Water and salt are conserved to $10^{-9}$ kg.

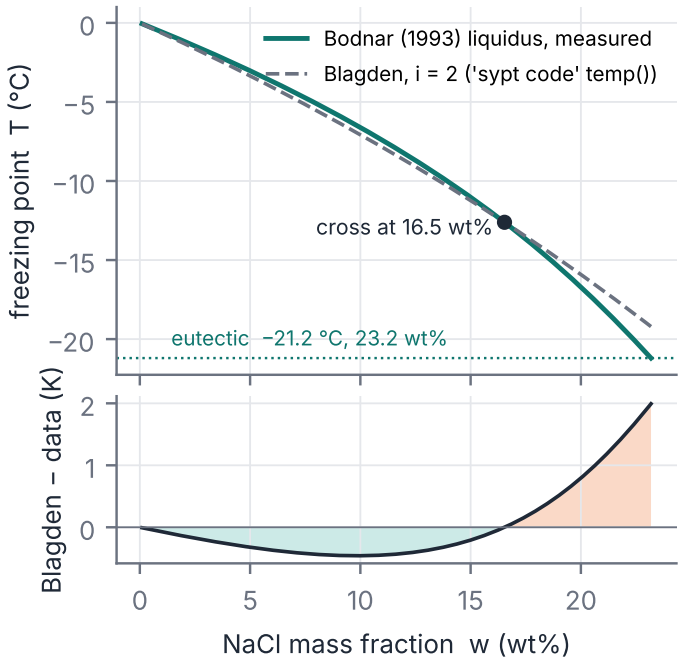



**Figure 1. FLOOR**. Freezing point against composition: Bodnar's measured liquidus and Blagden's law as coded in `temp()`. Lower panel: Blagden minus data. The residual is small for the dilute brines and grows to 2.0 K at the eutectic, where Blagden also loses the floor.

# Part II · What the models say

## 12 How cold: results of the equilibrium model

### 12.1 Blagden's law against the data

**FLOOR**. Blagden's law (10), as coded in `temp()`, crosses Bodnar's liquidus (11) at **16.5 wt %** and **−12.61** °C, the dossier's value to rounding. The ratio of ideal to measured depression is 1.106, 1.069, 1.019, 0.952, 0.909 at 5, 10, 15, 20, 23 wt % (Fig. 1). The crossing belongs to the linear approximation: keeping $\ln(1-x)$ in (9) instead of $-x$ moves it near 12 wt %.

### 12.2 Pre-dissolving has a floor near $-13\,^{\circ}\mathrm{C}$

Salt dissolved in water at room temperature spends its $\Delta h_s$ cooling water that the room then rewarms, and the brine arrives at the ice carrying sensible heat that can only be paid in meltwater, which dilutes it. For the five pre-dissolved recipes in the 10 Sep notes the model gives $-12.0$ to $-12.5\,^{\circ}\mathrm{C}$ (notes: $-12.1$ to $-12.6$); **saturating the brine at 25 °C moves the floor only to $-12.9$ to $-13.1\,^{\circ}\mathrm{C}$**. The 10 Sep conclusion — the eutectic needs staged addition onto ice — follows, and the staged protocol 20 + 6 + 6 + 6 g reproduces as $-12.4$, $-15.8$, $-19.1$, $-21.2\,^{\circ}\mathrm{C}$ (notes: $-12.4$, $-15.7$, $-18.9$, $-21.2$).

### 12.3 How much ice the eutectic needs

By (14), for 5.6 kg of water in total, 1.0 kg of salt and water at 4 °C, the ice fraction must exceed **0.612** to reach $T_E$ and **0.691** to keep 150 g of salt solid (ice at 0 °C); 0.555 and 0.627 with ice from a $-18\,^{\circ}\mathrm{C}$ freezer (Fig. 5).

**Table 2.** Reproduction of earlier B8 numbers. Each earlier number assumed a starting temperature it did not state; the column says which one reproduces it.

| Case | Notes | This model | Start |
|---|---|---|---|
| 75 g ice + 25 g water + 18 g salt | eutectic, ∼2 g solid | $-21.2\,^{\circ}$C, 2.7 g solid | 25 °C water |
| stall at $-7.5\,^{\circ}$C forces $w$ | 11.1 wt % | 11.1 wt % | – |
| 150 + 60 + 20 pre-dissolved | $-12.4\,^{\circ}$C | $-12.42\,^{\circ}$C | 25 °C brine |
| staged 20 + 6 + 6 + 6 g | $-12.4/-15.7/-18.9/-21.2$ | $-12.4/-15.8/-19.1/-21.2$ | 25 °C |
| 1200 ice + 400 water, 20 g salt | $-2.07\,^{\circ}$C | $-2.08\,^{\circ}$C | 25 °C water |
| … 90 g / 200 g | $-8.61$ / $-17.58\,^{\circ}$C | $-8.67$ / $-17.92\,^{\circ}$C | 25 °C water |
| Blagden crossover | 16.4–16.5 wt % | 16.5 wt % | – |
| drift, four bath sizes | $+5.91/1.86/0.97/0.49$ K | $+5.91/1.85/0.97/0.49$ K | on liquidus |
| $L_{\text{eff}}/L_f$ | 1.15–1.70 | 1.15 at $-11.2\,^{\circ}$C | – |
| latent share of the cold | ∼92 % | 87 % (0 °C) to 90–92 % (25 °C) | see note |

## 12.4 Drift and the buffer

On the liquidus, 300 g of salt at 20 wt % gives $+17.3$ K per kg melted by (13) (equation bank: $+17$). Rebuilding the rig note's four baths on their own liquidus and melting 150 g gives $+5.91$, $+1.85$, $+0.97$, $+0.49$ K, its table to two decimals, the $1/$size law. On the plateau, forming 1 kg of eutectic brine absorbs **241 kJ** and consumes 0.960 g of solid salt per kJ. The rig note's 272 kJ kg$^{-1}$ used $L_0$ at 0 °C; at $-21.2\,^{\circ}$C the latent heat is lower (6), so the buffer is used up **13 % faster** than that note assumed.

## 12.5 Rate and floor optimise at different loadings

**RATE** and **FLOOR** part company (Fig. 3b). For 4 kg ice and 1.6 kg water, the time to 5 °C falls with salt until the starting bath reaches $T_E$ at salt : ice $\approx 0.20$ (2.82 min) and is flat beyond. The drink temperature after 40 min keeps falling until $\approx 0.23$, where enough solid salt survives the drink and 40 min of leak to keep the bath pinned. Beyond that, more salt buys only hold time.

## 12.6 The earlier notes, reproduced

Where a number does not reproduce, the starting temperature is the reason. With the water at 0 °C instead of 25 °C the Session-1 sweep runs 0.6–3.3 K colder at every step, and the latent share drops to 87 %: warm free water must be paid for by melting ice, which raises the latent share. The earlier 92 % therefore belongs to a warm-water charge. The conclusion is unchanged — the salt supplies about a tenth of the cold and the ice the rest — but the earlier notes should state their starting temperatures.

## 12.7 Numbers for the rig

Tank 7.21 L, Can 2 (∅ 66.0 mm, 116.6 mm, $A = 310.2$ cm$^2$), 330 g of drink, EPS box at 0.46 W K$^{-1}$, room 28 °C, drink from 28 °C. Ice at 0 °C, water at 4 °C, salt at room temperature.

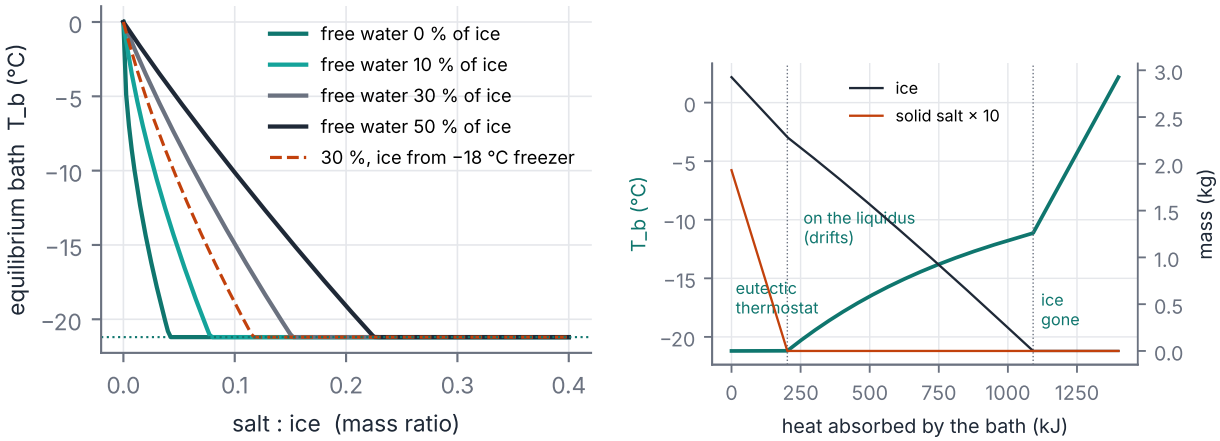



**Figure 2. FLOOR**. *Left:* adiabatic bath temperature against salt : ice, ice at 0 °C. More free water needs more salt to reach the floor; at 30 % free water, $-18\,^{\circ}$C freezer ice cuts the salt needed from 0.15 to 0.12. *Right:* the buffered charge (§12.7) as heat is added. Three regimes in order: the plateau, where solid salt is consumed at fixed $T_E$; the liquidus, where the bath drifts as ice melts; ice exhausted, where it warms freely.

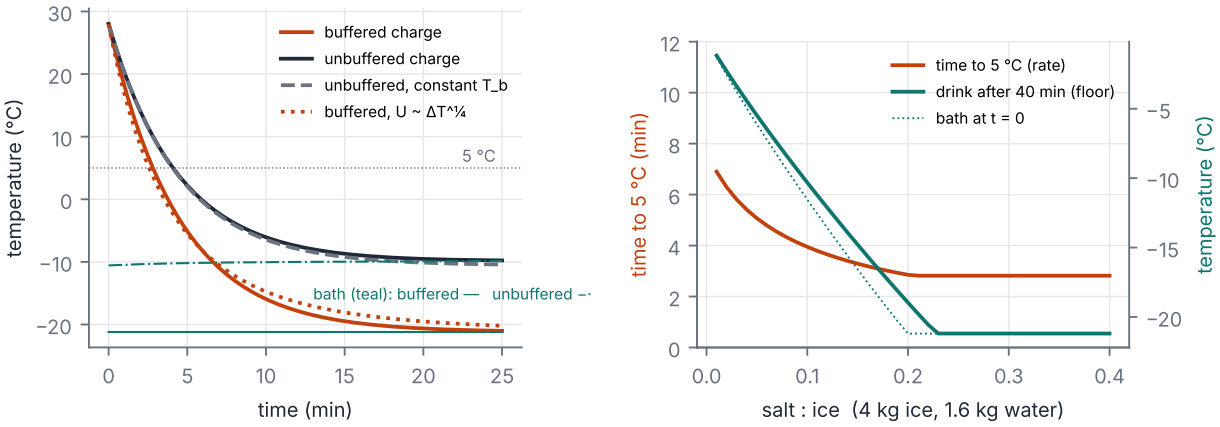



**Figure 3. RATE**. *Left:* Can 2 from 28 °C in three charges. The buffered bath holds $-21.2\,^{\circ}$C throughout; the unbuffered one warms by 0.7 K, and the constant-$T_b$ model misses that by 2.2 s in time to 5 °C. The dotted curve is the $\Delta T^{1/4}$ law. *Right:* time to 5 °C (rate) and drink temperature after 40 min (floor) against salt; they saturate at 0.20 and 0.23.

| | rig note<br>3.0 : 2.6 : 0.95 kg | **buffered**<br>**4.0 : 1.6 : 1.00 kg** | unbuffered<br>3.0 : 2.6 : 0.55 kg |
|---|---|---|---|
| starting $T_b$ | $-17.78\,^{\circ}$C | $-$**21.20** °C | $-10.57\,^{\circ}$C |
| regime | liquidus | eutectic | liquidus |
| solid salt at start | 0 g | 193 g | 0 g |
| time to 5 °C (Newton) | 189 s | 169 s | 246 s |
| $\Delta T^{1/4}$ law | – | 155 s | – |
| $T_b$ after 25 min | $-16.50\,^{\circ}$C | $-21.20\,^{\circ}$C | $-9.86\,^{\circ}$C |
| ice melted / salt used in 25 min | 222 g / – | 316 g / 95 g | 204 g / – |
| blank-run warming | 1.13 K h$^{-1}$ | 0 for 2.5 h | 0.62 K h$^{-1}$ |

**The rig-note charge has no thermostat.** It saturates its free water, which is what the rig note checked, but adiabatically it lands at $-17.8\,^{\circ}$C with every gram of salt dissolved, and even with $-18\,^{\circ}$C freezer ice only at $-19.5\,^{\circ}$C. The "$\geq 150$ g stays solid" line was a mass balance; the energy balance (14) says otherwise. **Use 4.0 kg ice + 1.6 kg water + 1.0 kg salt**: $-21.2\,^{\circ}$C with 193 g solid,

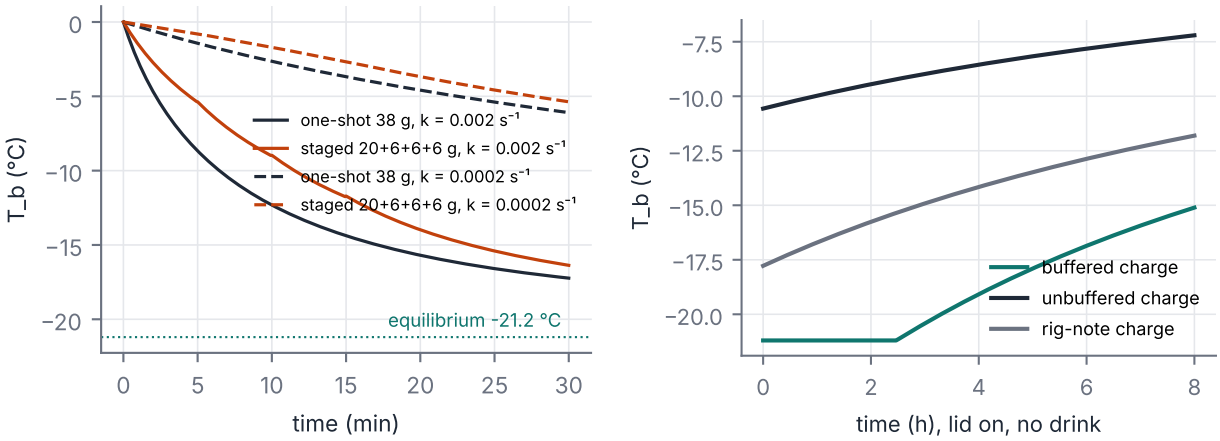



**Figure 4.** *Left:* well-mixed kinetic dissolution (41), 150 g ice + 60 g water, the same 38 g of salt added at once or in four tranches, for two rate constants. One-shot is *faster* in this model — the opposite of the bench (§14). *Right:* blank runs, lid on, no drink. The buffered charge holds the plateau for 2.5 h on the leak alone.

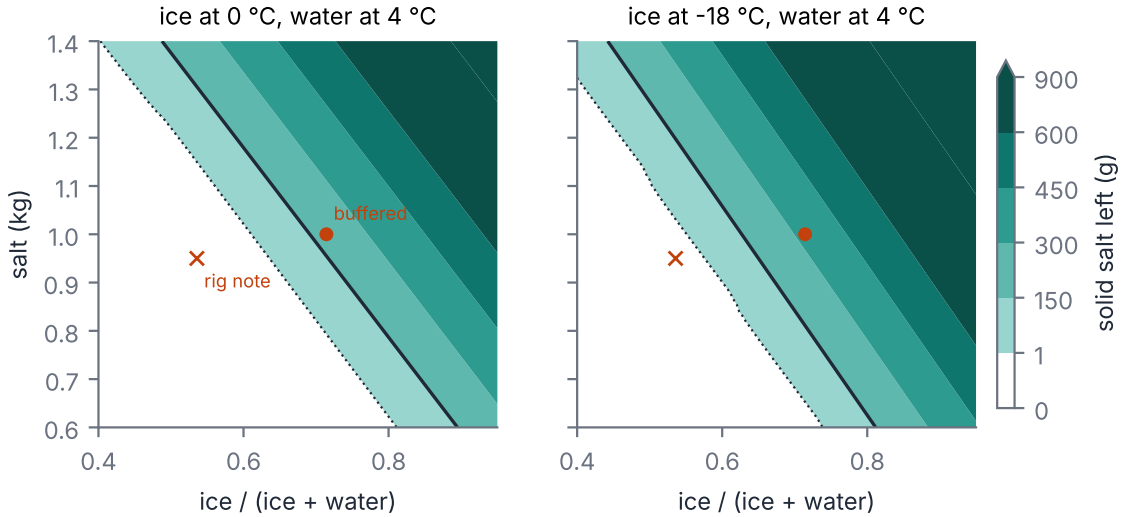



**Figure 5. FLOOR**. Solid salt left at adiabatic equilibrium, 5.6 kg of water in total, water at 4 °C, salt at 28 °C. White: never reaches $T_E$. Dotted: just reaches it. Solid line: 150 g of salt stays solid. Cross: the rig-note charge. Dot: the buffered charge of §12.7.

an equilibrium volume of about 6.2 L, plus 0.40 L for the can, inside the 6.8 L to the 16.5 cm line. Add the ice in two lots; 4 kg of crushed ice heaps above the brim until the first lot melts back. One drink plus 25 min of leak uses 95 g of the 193 g, so top up the salt between runs and watch for crystals on the floor.

**The blank-run inversion needs $L_{\text{eff}}$.** Inverting the warming rate as $\dot{Q} = L\,\dot{T}_b/(\mathrm{d}T_b/\mathrm{d}m)$ recovers the true 17.7 W leak of the unbuffered charge as 15.5 W with $L_f$ ($-13\,\%$) and 17.6 W with $L_{\text{eff}}$ (13) ($-1\,\%$); for the rig-note charge, 16.9 against 21.1 W ($-20\,\%$) and 20.8 W ($-1\,\%$). Fit the slope over the first ten minutes: $\mathrm{d}T_b/\mathrm{d}m$ itself falls as the bath dilutes, so a one-hour average biases the answer by another 5 %. And the buffered bath cannot be used for this at all — it reads zero slope for 2.5 h — which is why the rig note asked for the blank to be run unbuffered.

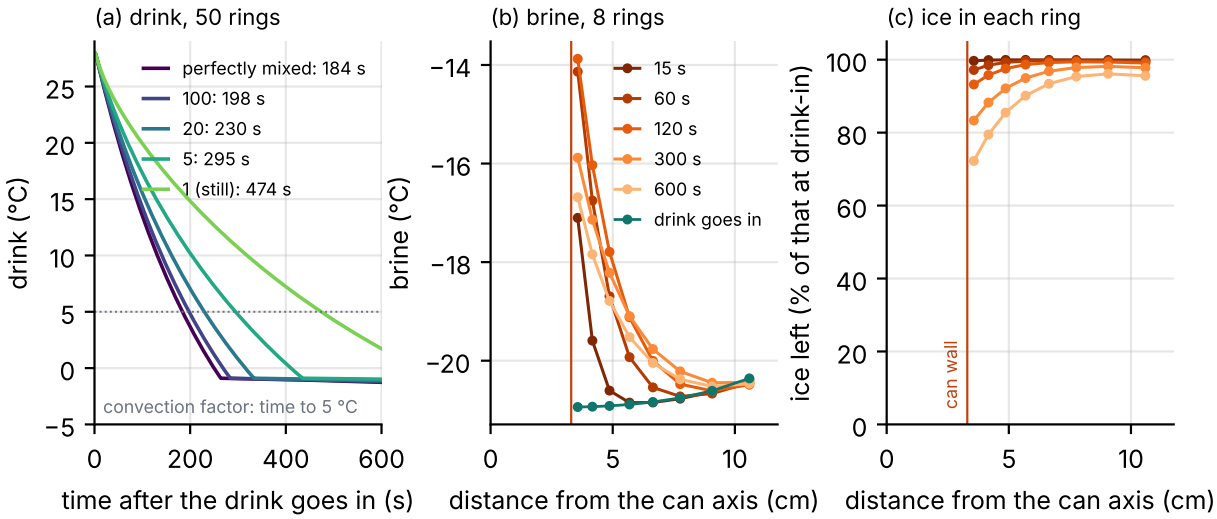



**Figure 6.** Heat spreading through the bath. (a) The drink for convection factors from perfectly mixed to still brine, run with 50 rings. (b) Brine temperature in each of the 8 rings, and (c) the share of each ring's ice still there, at times after the drink goes in; the default inputs. `python3 b8_simple.py --report-figure` redraws it.

## 13 How fast: results of the simulator

The simulator's numbers are for the buffered charge on the rig of 20 October, with every input in `inputs.toml`, the can lying on its side, and are printed by `python3 b8_simple.py`. With the bath perfectly mixed it gives 184 s to 5 °C, against 169 s in the table of §12.7, which had the whole outside of an upright can passing heat (§9.4).

With the defaults the drink reaches 5 °C in **222 s instead of 184 s** for a perfectly mixed bath. The inner ring warms 6 K above the rest within a minute and peaks at $-13.8\,^{\circ}\mathrm{C}$, while the bulk stays at $-20.4\,^{\circ}\mathrm{C}$; the ice next to the can goes first (Fig. 6c: more than a quarter of it in 10 minutes, against 4 % at the walls). Even before the drink, the bath is not uniform: the glass warms the outer ring, 0.3 K above a perfectly mixed bath. Over the run, 138.2 kJ comes from the drink and 43.2 kJ from the room through the box.

| $f$ | time to 5 °C | | brine by the can, peak | bath at drink-in |
|---|---|---|---|---|
| | 8 rings | 50 rings | 8 rings | 8 rings |
| $\infty$ (perfectly mixed) | 184 s | 184 s | $-19.6\,^{\circ}\mathrm{C}$ | $-20.70\,^{\circ}\mathrm{C}$ |
| 100 (stirred) | 196.5 s | 198 s | $-17.6\,^{\circ}\mathrm{C}$ | $-20.62\,^{\circ}\mathrm{C}$ |
| **20 (default)** | **222 s** | **230 s** | $\mathbf{-13.8}\,^{\circ}\mathrm{C}$ | $\mathbf{-20.36}\,^{\circ}\mathrm{C}$ |
| 5 | 266 s | 295 s | $-9.0\,^{\circ}\mathrm{C}$ | $-19.88\,^{\circ}\mathrm{C}$ |
| 1 (still brine) | 345.5 s | 474 s | $-3.5\,^{\circ}\mathrm{C}$ | $-19.30\,^{\circ}\mathrm{C}$ |

**Rings.** The rings need to resolve $\ell$ (28). At the default, 4, 8, 16, 32 and 50 rings give 214, 222, 226.5, 229 and 230 s, so 8 are within 4 %. In still brine $\ell$ is 0.6 cm and shrinks as the ice by the can melts away, so 8 rings (345.5 s) are far from the 50-ring 474 s; use 50 for $f$ below about 5. Between $f = 20$ and 60 the time to 5 °C moves only from 222 to 202.5 s, because the can's own films then dominate; below 10 it rises quickly.

**What matters most.** Over plausible ranges the inputs that move the time to 5 °C most are $h_o$ ($\div 3$ to $\times 3$: 442.5 to 147.5 s), a measured $U$ (120–200 W m$^{-2}$K$^{-1}$: 339.5 to 218.5 s), `on_side_factor` (1.0–

1.4: 259 to 195.5 s), the drink's mass or heat capacity ($\pm 20$ %: 268 to 176.5 s) and $f$ (5–100: 266 to 196.5 s); then the can's orientation (on its side or upright: 222 or 246 s) and the headspace (0.6–2.0 cm: 210 to 237 s). The **FLOOR** numbers hardly move. A fit of $U$ from a drink run (E3) depends on where the brine probe is: **within about 5 mm of the can it measures the can alone**; out in the bath it measures the can and the spreading together, and should be used with $f = \infty$. Two brine probes, one at the can and one at least 5 cm away, measure $f$: the default predicts a peak difference of about 6.5 K, still brine 16 K, a stirred bath 3 K.

**The box.** A box leaking anything from 0.25 to 1.0 W K$^{-1}$ gives 222 to 223 s: next to the can's 4.9 W K$^{-1}$ the leak hardly touches the drink. It shows in the bath instead: by the end of the 50-minute run the default bath is at $-19.3\,^{\circ}$C with 2.09 kg of ice left, against $-17.5\,^{\circ}$C and 1.93 kg at 1.0 W K$^{-1}$.

**Still left out.** Each ring has one temperature from floor to surface, so the vertical layering of §14 is not in it. Convection is a fixed multiple of $k_b$, not a flow driven by the temperatures it carries. The tank is taken as a cylinder of the same floor area, and the drink as one temperature. Crushed ice pressed against the can or the glass would conduct directly; only brine is counted.

**The simulator.** Every number it uses is in `inputs.toml`, with its unit and a tag saying where it comes from; a short file with only the lines that change overrides it. `b8_app.py` and the web page (`https://darkpyxlz.github.io/sypt/`, the same Python files run in the browser) put the main inputs on sliders, each with a box for typing an exact value, and draw the tank to scale with the brine ring by ring. A run takes about 0.5 s in Python and 1 s in the browser.

# Part III · What to do next

## 14 Open problem: why did the one-shot bath stall?

On 10 Sep, 75 g ice + 25 g water + 18 g salt added at once stalled at $-7.5\,^{\circ}$C, against an equilibrium of $-21.2\,^{\circ}$C. The notes attribute this to a crust on the floor in cold, near-saturated brine. The simulation can test the simplest version of that idea, a well-mixed bath with the dissolution law (26) written as

$$\dot{s} = k\,(S - s)^{2/3}\,S^{1/3}\,\frac{w_{\mathrm{sat}}(T_b) - w}{w_{\mathrm{sat}}(T_b)}, \tag{41}$$

*Where it comes from.* This is (26) with the grain area proportional to the salt put in. The rate constant $k = k_d\,\rho_b\,w_{\mathrm{sat}}\,A_{s0}/S$, in s$^{-1}$, gathers the constants. Whatever $k$ is, **this law makes the one-shot bath faster than the staged one** (Fig. 4): at 10 min, $-12.3$ against $-9.0\,^{\circ}$C at $k = 2\times10^{-3}$ s$^{-1}$, because the one-shot bath has more crystal area exposed from the start. The bulk brine at $-7.5\,^{\circ}$C is only 11 wt % against a saturation of about 25 wt %, so in a mixed bath the drive never vanishes and nothing can stall.

So the stall is evidence that the bath was not well mixed. Dense saturated brine around the crust is gravitationally stable and stays on the floor; the crust then dissolves into a layer that is already saturated, and the bulk only receives salt as fast as that layer is exchanged. Stirring sets the exchange rate, and its absence sets it near zero. If this is right, the result is not a dissolution constant but a mixing one, and it predicts something cheap to check: **a probe on the floor of a stalled one-shot bath should read colder than the bulk**, near the liquidus of saturated brine, and stirring should release the stall in seconds. A probe reading the same as the bulk kills the stratification picture and puts the explanation back in surface kinetics.

# 15 Experimental programme

**E1 · Charge check, one afternoon.** Mix the rig-note and buffered charges side by side in the tank, logging $T_b$ until it settles. Predicted $-17.8$ and $-21.2\,^{\circ}$C, with crystals on the floor of the second only. IV: charge recipe. DV: settled $T_b$, presence of solid salt. Controls: starting temperatures of all three components, recorded.

**E2 · The box's leak.** Unbuffered charge, lid on, no drink, one hour. Fit $\dot{T}_b$ over the first ten minutes; invert with $L_{\text{eff}}$ (13). Predicted 0.62 K h$^{-1}$ for 0.46 W K$^{-1}$ with the stirrer off; the stirrer's power adds to the leak, so measure it once with the stirrer off as well. DV: $(UA)_{\text{box}}$. A simpler measurement of the same number, with no phase change to invert, is the box test of §10.3: cold water only, three hours, (39).

**E3 · Drink runs.** Can 2 from room temperature in the buffered and unbuffered charges, lying on its side as in the rig. DV: $T_d(t)$, $T_b(t)$. Fit both Newton and the $\Delta T^{1/4}$ law to each run, compare residuals, and report $U$ rather than $\tau$. Log the brine with two probes, one within 5 mm of the can and one at least 5 cm away: fitted against the first, $U$ is the can's alone; the difference between them measures the convection factor $f$ (§8.3), predicted to peak near 6.5 K. Feed the measured $T_b(t)$ of the unbuffered run into (22) as a check of the bath model independent of the drink model.

**E4 · Stall test.** 150 g ice + 60 g water + 38 g salt at once, unstirred, probes at the floor and mid-depth. Then repeat staged. DV: floor minus bulk temperature; time for the stall to clear after stirring starts.

**E5 · The original charge.** 1 kg ice, 2 kg water at 25 °C, 0.5 kg salt. Predicted $-10.5\,^{\circ}$C with about 50 g of ice left, so the ice should be visibly almost gone. A direct, memorable test of why `temp()` could not work.

Suggested order: E1, E2, E3, E4, E5. E1 decides whether E3 has a thermostat.

# 16 Estimates justifying the approximations

## 16.1 One bath temperature

`b8_sim.py` gives the bath a single $T_b$; the simulator splits it into rings round the can (§8.4), with the brine by the can 6.5 K above the bulk at the peak. Neither resolves height: §14 argues that layering is what stalls an unstirred one-shot bath. With the stirrer on and probes at least a few centimetres from the can, a single $T_b$ is the right first model for the **FLOOR**.

## 16.2 The drink as a lump

The drink's Biot number is about 9 (§6.6), so it is not a lump by conduction; internal convection rescues it by a factor of about 9. The lumped model is the volume average of the PDE solved in COMSOL, which is the justification, and the 4 °C density maximum of water is outside it.

### 16.3 The eutectic solid

At $T_E$ the equilibrium solid is ice + hydrohalite NaCl·2H$_2$O, not ice + halite. The formation enthalpy is $-0.73\,\mathrm{kJ\,mol^{-1}}$ (Drebushchak 2017), 12.5 kJ per kg of salt, against about 1040 kJ per kg of salt consumed on the plateau: 1.2 %. The water locked into hydrohalite matters more for the buffer accounting, but hydrohalite nucleates reluctantly, so on the time scale of one run the halite model is the better description.

### 16.4 Assumed constants

Changing each placeholder over a plausible range (`run_checks()` section m) moves the unbuffered $T_b$ by at most 0.11 K, the rig-note $T_b$ by at most 0.29 K, and the buffer by at most 16 g of its 193 g. No **FLOOR** conclusion depends on them.

| Change | $\delta T_b$ unbuffered | $\delta$ solid salt, buffered | $\delta T_b$ rig note |
|---|---|---|---|
| $\Delta h_s \times 0.7$ / $\times 1.3$ | $+0.11$ / $-0.11$ K | $-15.7$ / $+15.1$ g | $+0.28$ / $-0.29$ K |
| $c_i = 2.10$ / 1.94 | $+0.02$ / $-0.01$ K | $-5.9$ / $+3.6$ g | $+0.05$ / $-0.03$ K |
| $c_b \mp 0.2$ | $-0.08$ / $+0.08$ K | $+13.9$ / $-14.4$ g | $-0.25$ / $+0.24$ K |

### 16.5 Numerics

Against a constant bath the equilibrium model's integrator reproduces the time to 5 °C of (23) to 0.01 s (169.07 s both), and that of the power-law closed form to 0.01 s (154.61 s both). Quartering the step changes $t_{5\,^\circ\mathrm{C}}$ by $10^{-3}$ s. The heat released by the drink and the time integral of the flux agree to $10^{-3}$ kJ in 65.44 kJ. The simulator closes its energy bookkeeping to $10^{-9}$ J (§11).

### 16.6 What the can's $U$ does

$U$ is the least certain number in the model. Time to target scales as $1/U$ (23); the **FLOOR** results do not depend on it at all. The SINTEF benchmark it came from cannot also be used to validate it. A $U$ fitted against a probe out in the bath includes the spreading of heat through the bath; against a probe at the can wall it does not.

## 17 Literature

Marked ◐ where only a secondary or abstract-level source has been read.

*Thermodynamics of the bath.* Atkins, P. & de Paula, J., *Physical Chemistry* (Oxford) — chemical potential, the colligative laws and the phase rule of §4. Bodnar, R. J., *Geochim. Cosmochim. Acta* **57**(3), 683–684 (1993) — the liquidus, (11) ◐ (equation via the Lille fluid-inclusion resource). Swenne (1983), PhD thesis, TU Eindhoven — $T_E = 252.03$ K, peritectic. Drebushchak, Ogienko & Yunoshev, *Thermochim. Acta* **647**, 94–100 (2017) — metastable eutectic near $-27\,^\circ$C, hydrohalite formation enthalpy. IAPWS R10-06 (2009), Revised Release on the Equation of State 2006 for H$_2$O Ice Ih — $c_i(T)$. Melinder, Å., *Properties of Secondary Working Fluids for Indirect Systems*, 2nd ed., IIR (2010) — $c_b$, $\rho_b$ and $k_b$ of NaCl brine, via CoolProp (INCOMP::MNA) ◐. Sigma-Aldrich solubility table — the fit in `sypt code`. Purdue thermochemical table — $\Delta h_s$ ◐.

*Heat and mass transfer.* Incropera, F. P. et al., *Fundamentals of Heat and Mass Transfer* (Wiley) — conduction, resistances, film coefficients and radiation (§6). Bejan, A., *Convection Heat Transfer* (Wiley) — the scaling of (20). Churchill, S. W. & Chu, H. H. S., *Int. J. Heat Mass Transfer* **18**, 1323–1329

(1975) — free convection on vertical walls and horizontal cylinders. Chem. Eng. Educ. beverage-bottle study (WPI) — unstirred $h_o$, $h_i$, wall Biot numbers. SINTEF can data — the $\tau$ behind $U_0$. Schulze Grachtrup, arXiv:1702.07014 (2017) — bottle benchmarks. Holland, D. M. & Jenkins, A., *J. Phys. Oceanogr.* **29**, 1787–1800 (1999) — the three-equation ice/brine interface (25) ◐. Nield, D. A. & Bejan, A., *Convection in Porous Media* (Springer), with Cheng, P. & Minkowycz, W. J., *J. Geophys. Res.* **82**, 2040–2044 (1977) — free convection in a porous bed (30) ◐. Carman, P. C., *Trans. Inst. Chem. Eng.* **15**, 150–166 (1937) — the Kozeny–Carman permeability ◐. Linden, P. F., *Annu. Rev. Fluid Mech.* **31**, 201–238 (1999) — exchange flow through openings (38) ◐.

*Mixing and dissolution.* Koefod (2017) — mixing raises the dissolved share of salt, supporting §14 ◐. Xu, Bootsma, Verzicco, Lohse & Huisman, *J. Fluid Mech.* **1019**, A11 (2025), and Yang, Howland, Liu, Verzicco & Lohse, *J. Fluid Mech.* **969**, R2 (2023) — mechanism only; both lie an order of magnitude below the bath's salinity. Pyodide (`https://pyodide.org`) — Python in the browser, for the web simulator.

## 18 Notes for whoever picks this up

**Do first.** E1. It costs one afternoon and decides between $-17.8$ and $-21.2\,^{\circ}$C for every later run, and whether the bath in E3 is a thermostat or a drifting reservoir.

**Do not** quote the rig-note charge as buffered; do not invert a blank run with $L_f$; do not report a single latent-heat share without the starting temperature of the water; and do not fit $U$ against a bulk probe and then use it with heat spreading switched on.

**The sharpest predictions**, cheapest first: the rig-note charge settles near $-17.8\,^{\circ}$C with no crystals left; the original `sypt code` charge leaves only about 50 g of ice; the buffered bath holds $-21.2\,^{\circ}$C for about 2.5 h on the leak alone; the brine at the can runs 6.5 K above the bulk; a stalled one-shot bath is colder at the floor than in the bulk.

**Software.** `b8_sim.py`: Python 3 with NumPy and SciPy, Matplotlib for the figures; everything runs in about 12 s. The simulator (`b8_model.py`) needs only Python 3.11 or later, plus Matplotlib for its pictures; one run takes about 0.5 s.

# A Symbols

| Symbol | Unit | Meaning |
|---|---|---|
| $T$, $T_b$, $T_d$, $T_s$ | °C | temperature; of the bath (brine), the drink, the ice surface |
| $H$, $h$ | J, J kg$^{-1}$ | enthalpy, and per kilogram |
| $W$, $m_i$, $m_l$ | kg | all water; ice; liquid water in the brine |
| $S$, $s$ | kg | all salt; dissolved salt |
| $w$, $w_L$, $w_{\text{sat}}$, $r$ | – | salt mass fraction of brine; on the liquidus; at saturation; $w/(1-w)$ |
| $T_E$, $w_E$; $T_P$, $w_P$ | °C, – | eutectic; peritectic |
| $c_w$, $c_i$, $c_s$, $c_b$ | J kg$^{-1}$K$^{-1}$ | heat capacity of water, ice, salt, brine |
| $L_0$, $L(T)$, $L_{\text{eff}}$ | J kg$^{-1}$ | latent heat at 0 °C, at $T$, as the bath charges it |
| $\Delta h_s$ | J kg$^{-1}$ | heat of solution of NaCl |
| $\mu$, $a_w$, $K_f$, $b$, $i$ | – | chemical potential, water activity, cryoscopic constant, molality, ions per formula |
| $k$, $\rho$, $\alpha$ | W m$^{-1}$K$^{-1}$, kg m$^{-3}$, m$^2$ s$^{-1}$ | conductivity, density, diffusivity |
| $\nu$, $\beta$, $g$ | m$^2$ s$^{-1}$, K$^{-1}$, m s$^{-2}$ | kinematic viscosity, expansion coefficient, gravity |
| $h$, $U$, $G$ | W m$^{-2}$K$^{-1}$, W m$^{-2}$K$^{-1}$, W K$^{-1}$ | film coefficient, overall coefficient, conductance |
| Ra, Nu, Pr, Bi | – | Rayleigh, Nusselt, Prandtl, Biot numbers |
| $C_d$, $A$, $\tau$ | J K$^{-1}$, m$^2$, s | drink heat capacity, can area, cooling time constant |
| $r_c$, $H_c$, $h_s$, $h_g$ | m | can radius, length, headspace, gas strip depth |
| $A_f$, $Z$, $\varepsilon$, $t_h$ | m$^2$, m, –, m | tank floor area, liquid level, heap porosity, heap thickness |
| $f$, $k_{\text{eff}}$, $\ell$ | –, W m$^{-1}$K$^{-1}$, m | convection factor, effective conductivity, penetration length |
| $N$, $r_j$, $\psi_j$, $\lambda$ | –, m, –, m | rings, ring edges, ring shares, shell length |
| $k_d$, $A_s$, $D$ | m s$^{-1}$, m$^2$, m$^2$ s$^{-1}$ | dissolving coefficient, grain area, salt diffusivity |
| $\varphi$, $T_{f0}$ | –, °C | frozen fraction of the drink's water, its first freezing point |

# B Using the code

```
python3 b8_sim.py           # checks, then figs/*.pdf and numbers.json
python3 b8_sim.py --quick   # checks only
python3 b8_simple.py        # the simulator, one run: table, summary, picture
python3 b8_app.py           # the simulator behind sliders, each with a box to type into
python3 b8_simple.py --report-figure   # redraws figs/f9_spreading.pdf
python3 test_b8.py          # the simulator's checks
```

```
import b8_sim as B
B.mix(1.0, 2.0, 0.5, T_water=25)        # adiabatic mix -> dict(T, m_ice, w, s_solid, regime)
B.mix_charge(B.BUFFERED)                # a named rig charge
r = B.run_charge(B.UNBUFFERED, t_end=1500, drink=dict(law='power'))
B.time_to(r, 5.0)                       # seconds to 5 C
B.simulate(0.15, 0.06, 0.038, with_drink=False, k_diss=2e-3)   # kinetic dissolution
```

Constants marked `ASSUMED` in the source are the ones to replace with measurements: $c_i$, $c_b(w)$, $\Delta h_s$, the hydrohalite solubility line, the drink's $c$ and can mass, and $U_0$. To use a measured bath trace instead of the bath model, set `constant_bath=True` and replace the line that sets `T_b` in `simulate()` with an interpolation of the log.

The heart of the equilibrium model, (7) and the solve of §5:

```python
def enthalpy(T, m_ice, m_liq_water, s_dissolved, s_solid):
    """Total enthalpy (kJ) of a bath at a single temperature T."""
    m_brine = m_liq_water + s_dissolved
    w = s_dissolved / m_brine if m_brine > 0 else 0.0
    return (m_brine * c_brine(w) * T          # sensible heat of the brine
            + s_dissolved * DH_SOL             # heat absorbed dissolving the salt
            + m_ice * (C_ICE * T - L_F)        # ice: sensible heat, minus latent heat
            + s_solid * C_SALT * T)            # undissolved salt


def _phases_at(T, W, s_cap, S):
    """Equilibrium phase amounts at a temperature strictly above T_E.
    W = all water (ice + liquid), s_cap = salt allowed to dissolve,
    S = all salt.  Returns (m_ice, m_liq_water, s_dissolved, s_solid)."""
    wl = w_liquidus(T)
    s_noice = min(s_cap, w_saturation(T) / (1.0 - w_saturation(T)) * W)
    w_full = s_noice / (s_noice + W)
    if T >= 0.0 or w_full >= wl:
        # Above the liquidus of the fully melted bath: no ice.
        return 0.0, W, s_noice, S - s_noice
    # On the liquidus: the brine must have composition wl,
    # and all available salt is in it (liquidus lies below saturation).
    m_lw = s_cap * (1.0 - wl) / wl
    return W - m_lw, m_lw, s_cap, S - s_cap


def bath_state(H, W, S, s_cap=None):
    """THE solve.  Given the bath's enthalpy H (kJ), total water W (kg),
    total salt S (kg), and how much salt may dissolve (s_cap, default all),
    return the equilibrium state as a dict.

    Four regimes, in order of decreasing H:
      'liquid'     no ice; T free                    (ice exhausted)
      'liquidus'   ice + brine on the liquidus; T = T_liq(w)
      'eutectic'   ice + brine + solid salt; T = T_E fixed   (F = C - P + 1 = 0)
      'solid'      below the eutectic, no liquid     (not reached in practice)
    """
    if s_cap is None:
        s_cap = S
    s_cap = max(0.0, min(s_cap, S))

    def pack(T, mi, mlw, s, ss, regime):
        mb = mlw + s
        return dict(T=T, m_ice=mi, m_liq_water=mlw, s_dissolved=s, s_solid=ss,
                    w=(s / mb if mb > 0 else 0.0), regime=regime)

    # --- pure water: the only invariant point is 0 C -----------------------
    if s_cap < 1e-9:
        if H > 0.0:
            T = H / (W * C_W + S * C_SALT)
            return pack(T, 0.0, W, 0.0, S, 'liquid')
        if H >= -W * L_F:
            mi = -H / L_F
            return pack(0.0, mi, W - mi, 0.0, S, 'liquidus')
        T = (H + W * L_F) / (W * C_ICE + S * C_SALT)
        return pack(T, W, 0.0, 0.0, S, 'solid')

    # --- the eutectic plateau: T fixed, enthalpy linear in brine mass -----
    r_E = W_E / (1.0 - W_E)                       # kg salt per kg water in eutectic brine
```

```
    m_max = min(W, s_cap / r_E)                  # most water that can be eutectic brine

    def H_eut(m_lw):
        s = r_E * m_lw
        return enthalpy(T_E, W - m_lw, m_lw, s, S - s)

    H_top, H_bot = H_eut(m_max), H_eut(0.0)

    if H >= H_top:                               # above the plateau: monotonic in T
        f = lambda T: enthalpy(T, *_phases_at(T, W, s_cap, S)) - H
        T = brentq(f, T_E + 1e-9, 200.0, xtol=1e-10)
        mi, mlw, s, ss = _phases_at(T, W, s_cap, S)
        return pack(T, mi, mlw, s, ss, 'liquid' if mi == 0.0 else 'liquidus')
    if H >= H_bot:                               # on the plateau
        m_lw = m_max * (H - H_bot) / (H_top - H_bot)
        s = r_E * m_lw
        return pack(T_E, W - m_lw, m_lw, s, S - s, 'eutectic')
    T = T_E + (H - H_bot) / (W * C_ICE + S * C_SALT)   # below the eutectic
    return pack(T, W, 0.0, 0.0, S, 'solid')
```